TECHGIUM OFFICIAL PROBLEM STATEMENTS

L&T; Technology Services Engineering Innovation Challenge Compendium

60 Problem Statements Across All Competition Tracks • Generated: September 23, 2026

Engineering Track / Domain	Challenge Count	% of Total
Manufacturing	20	33.3%
MedTech	10	16.7%
AI	9	15.0%
Energy	5	8.3%
Mobility	4	6.7%
Digital Platforms	4	6.7%
Cyber Security	4	6.7%
Robotics & Autonomous Systems	2	3.3%
Sustainability	2	3.3%
Total Official Problem Statements	**60**	**100.0%**

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PROBLEM STATEMENT #1 • REF: 6a996c5e239fdeda MANUFACTURING

AI enabled Conceptual Design & Costing for Pharmaceutical Clients

1. Background & Industrial Context:

Conceptual planning for pharmaceutical and biopharmaceutical facilities requires coordinated decisions across process engineering, architecture, equipment, utilities, automation, safety, validation, scheduling, and commercial functions. Teams must translate incomplete or changing client requirements into workable room arrangements, equipment locations, personnel and material flows, utility needs, and preliminary investment estimates. Today, this work depends heavily on spreadsheets, reference projects, manual calculations, and expertise. Design assumptions may be recorded inconsistently, making alternatives difficult to compare and decisions difficult to trace. Changes in capacity, equipment, zoning, or compliance requirements can trigger repeated layout and costing revisions across disciplines. These challenges are particularly significant during early bids, when information and time are limited but reviewers still need confidence that the proposed concept is feasible, compliant, and commercially credible. The core need is a consistent method for capturing inputs, applying approved engineering knowledge, comparing options, and presenting assumptions, constraints, risks, costs, and decisions clearly for expert review.

2. Challenge Statement:

Create an AI-assisted proof of concept for one pharmaceutical facility area, such as a production block or utility zone. The POC should accept a structured brief containing room requirements, major equipment, capacities, process relationships, material and personnel flows, selected design rules, and cost parameters. It should generate two or three conceptual layout alternatives, rank them using transparent criteria, and explain key trade-offs. The selected alternative must be represented in a lightweight 3D Building Information Modeling (BIM) model showing rooms, major equipment, access, and spatial relationships. The system should perform basic rule checks for zoning, adjacency, flow separation, safety, maintainability, and space allocation, then produce an indicative Total Investment Cost range with traceable assumptions. Use synthetic, sanitized, or mentor-approved reference data.

3. Expected Deliverables & Outcomes:

• Structured input template and documented assumptions for one selected facility area.
• Two or three conceptual layout alternatives with transparent scoring and trade-off comparison.
• Lightweight 3D BIM model for the selected alternative.
• Rule-check report covering zoning, adjacency, flows, safety, maintainability, and space allocation.
• Indicative Total Investment Cost range with cost categories, sources, assumptions, and exclusions.
• Simple option-comparison dashboard and validation against one mentor-approved reference case.
• Working POC, source code, test evidence, user guide, technical documentation, and future-enhancement recommendations.

PROBLEM STATEMENT #2 • REF: 6a996c5e239fdeda MANUFACTURING

Leveraging AI in CAD cleansing activities

1. Background & Industrial Context:

CAD models move through design, review, digital mock-up, release, manufacturing planning, supplier exchange, and product-lifecycle systems. Before reuse, they may require cleansing to correct inconsistent naming, missing metadata, broken references, duplicated or hidden geometry, nonstandard layers, unresolved features, invalid structure, excessive detail, or compliance issues. Required checks and acceptable corrections differ by CAD tool, product type, customer standard, lifecycle stage, and downstream use. Engineers often interpret check reports, consult procedures, inspect models manually, and decide which findings can be corrected without changing design intent. This work can be repetitive and difficult to estimate, yet incorrect automated changes may damage geometry, remove information, or create downstream problems. Limited labeled examples and proprietary file formats also make generalized automation difficult. The core need is a trustworthy approach that identifies relevant quality issues, explains their impact, supports consistent remediation, and preserves design intent while adapting to different standards, data availability, and engineering workflows.

2. Challenge Statement:

Demonstrate a proof of concept that improves a defined CAD cleansing activity or workflow. Teams may choose the CAD representation, model type, quality problems, customer rules, lifecycle stage, and downstream use. The solution should analyze sample models or extracted information, identify or prioritize issues, and support correction, guidance, or review. Participants may combine configurable checks, geometric analysis, metadata reasoning, artificial intelligence, automation, human-in-the-loop interaction, or other justified approaches. The POC should show how rules or knowledge can be adapted for more than one scenario without claiming universal support for all CAD systems. It must distinguish safe automatic actions from changes requiring engineer approval, retain traceability to the original model, and provide evidence that design intent is not altered. Teams should define success measures such as detection quality, correction accuracy, effort reduction, consistency, or downstream readiness.

3. Expected Deliverables & Outcomes:

- Clearly defined CAD cleansing problem, users, model scope, standards, and success criteria.
- Representative test models or extracted CAD information with documented issues.
- Working POC for detecting, prioritizing, correcting, or guiding review of selected issues.
- Configurable or adaptable handling of at least two relevant cleansing scenarios.
- Human-review mechanism for uncertain or design-sensitive changes.
- Traceability showing findings, actions, approvals, and original-model preservation.
- Evaluation against expert-reviewed results or another justified baseline.

PROBLEM STATEMENT #3 • REF: 6a996c5e239fdeda MANUFACTURING

Non-Contact Crack Detection and Digital Twin Generation for Dies and Moulds

1. Background & Industrial Context:

Dies, moulds, and other high-value industrial tooling components are exposed to repeated thermal and mechanical loading during operation, resulting in progressive surface degradation and crack formation. Early identification and localization of such defects are critical to preventing unexpected failures, reducing maintenance costs, and extending component life. Traditional inspection techniques often rely on manual assessment, which can be time-consuming, subjective, and difficult to scale.

There is a growing need for intelligent inspection systems capable of autonomously detecting, mapping, and tracking surface defects with high accuracy. Such systems can serve as a foundation for predictive maintenance, digital inspection records, and automated repair processes. An effective solution should be capable of generating accurate spatial representations of defects and enabling seamless integration with downstream manufacturing or maintenance workflows.

2. Challenge Statement:

A XY positioning system tracks each crack location while the next crack location is being captured. The solution should identify defect geometry, generate an accurate spatial profile, and create a digital representation that can be used for inspection, analytics, maintenance planning, or automated repair applications. The proposed system may leverage computer vision, advanced sensing technologies such as laser, artificial intelligence, image processing, robotics, motion control, or a combination of these approaches. The solution should demonstrate the ability to accurately capture defect characteristics and support traceability through digital coordinate extraction or digital twin generation.

3. Expected Deliverables & Outcomes:

Take a metal plate and machine a random curve of 300µm width on it. This represents the crack.

Run the crack detection system that determines the crack while tracking the points using an XY positioning system. Record the coordinates traced by the XY positioning system in the form of a XY point cloud.

Use the XY point cloud in a CAD software to make a digital copy of the traced path. Print this on a transparent paper and overlay the paper on top of the crack.

The largest deviation between the crack and the overlayed print determines the accuracy of this sytem. The target accuracy is 300 µm or better.

The solution should integrate sensing and motion control to enable accurate crack localization, coordinate extraction, and automated tracing.

The report should include detailed explanations, diagrams, and calculations to showcase the design

PROBLEM STATEMENT #4 • REF: 6a996c5f239fdeda MEDTECH

Novel pinless/markerless registration methods for navigation and surgical robotics

1. Background & Industrial Context:

Surgical navigation depends on accurately aligning preoperative three-dimensional images and plans with the patient's anatomy in the operating environment. Current orthopedic workflows commonly use tracked pointers, reflective markers, rigid arrays, or bone-fixed pins to establish and maintain this relationship. These methods can add setup time, require repeated landmark collection, depend on operator technique, obstruct the working area, and introduce extra hardware or invasive fixation. Registration accuracy may also change when the anatomy, camera, or reference array moves, making continuous verification important. A markerless approach could simplify workflow, but it must address limited surface visibility, soft-tissue occlusion, similar anatomical shapes, sensor noise, viewpoint changes, and differences between preoperative images and intraoperative observations. The central engineering need is a noninvasive method that identifies observable bony geometry, aligns it with a CT-derived model, reports registration confidence, detects loss of alignment, and supports repeatable validation without relying on implanted fiducials or manual point correspondence.

2. Challenge Statement:

Develop a markerless registration proof of concept for one orthopedic anatomy and workflow, using a synthetic bone model, physical phantom, or approved dataset. The POC should create a three-dimensional reference model from CT or equivalent imaging and acquire intraoperative-like surface or depth observations using stereo vision, structured light, depth sensing, or another justified modality. It should automatically identify usable geometric features, estimate the rigid transformation between observed anatomy and the reference model, and display the aligned result. The system must calculate registration error and confidence, detect insufficient coverage or loss of alignment, and allow re-registration when required. Evaluation should include different viewpoints, partial occlusion, sensor noise, and controlled phantom movement. Teams should compare the method with a point-based or surface-based baseline. Instrument tracking can be demonstrated as an extension.

3. Expected Deliverables & Outcomes:

- Defined anatomy, workflow, sensing method, assumptions, and registration architecture.
- CT-derived or equivalent three-dimensional reference model and phantom or synthetic test dataset.
- Markerless surface or depth acquisition and preprocessing pipeline.
- Automatic feature extraction and rigid-registration algorithm with confidence reporting.
- Visualization showing reference, observed anatomy, alignment, and registration status.
- Detection of inadequate coverage, alignment loss, and re-registration need.
- Validation report covering target registration error, repeatability, occlusion, noise, movement, and baseline comparison.

PROBLEM STATEMENT #5 • REF: 6a996c5f239fdeda MEDTECH

Patient Specific 3D Anatomical Reconstruction from DICOM Imaging Data

1. Background & Industrial Context:

Medical imaging technologies such as CT, MRI, X-ray generate a large amount of anatomical information in the form of DICOM image slices. In most cases, these images are still reviewed as a sequence of 2D slices and understanding the complete 3D anatomical structure depends heavily on the experience of the clinician or researcher. When the anatomical region is complex, such as the heart, brain or breast, it becomes difficult to accurately understand the spatial relationship between different structures only from individual image slices.

Another challenge is that DICOM images may have variations in slice thickness, spacing, patient orientation, image quality and acquisition parameters. Patient movement and imaging artifacts can also introduce misalignment between slices. These variations can affect the accuracy of 3D reconstruction and subsequent anatomical analysis.

There is therefore a need to develop an approach that can process DICOM image data, maintain the correct anatomical and spatial relationship between image slices, reconstruct the region into an accurate 3D representation, and identify individual anatomical structures within it. The reconstructed model should also support detailed visualization and analysis of structural and functional characteristics, so that it can be useful for medical research, diagnostic assessment, treatment planning and surgical planning.

2. Challenge Statement:

The challenge is to develop a reliable method for reconstructing a patient-specific 3D anatomical model from medical imaging data available in DICOM format. The solution should be capable of reading a sequence of DICOM image slices, interpreting the associated metadata, and maintaining the correct spatial relationship between individual slices during reconstruction. Variations in slice orientation, spacing, thickness, image resolution, patient movement and acquisition quality should be appropriately considered so that they do not significantly affect the final anatomical representation.

The reconstructed 3D volume should support identification and separation of relevant anatomical structures present within the selected region. The solution should also be capable of refining the reconstructed structures and generating an anatomically meaningful 3D mesh that closely represents the original patient anatomy. Particular importance should be given to alignment accuracy, structural continuity, anatomical consistency and preservation of clinically relevant features.

The developed approach should further provide an interactive visualization of the reconstructed anatomical model and enable the analysis of important structural or tissue-related characteristics wherever applicable. The overall objective is to achieve a consistent, accurate and usable 3D anatomical representation that can support medical visualization, research analysis, diagnostic assessment, treatment planning and surgical planning.

3. Expected Deliverables & Outcomes:

The expected deliverables should include a working prototype capable of processing DICOM image data and reconstructing a patient specific 3D anatomical model. The solution should demonstrate DICOM metadata handling, slice ordering and alignment, 3D volume reconstruction, anatomical structure segmentation, mesh generation and validation. Participants should provide an interactive visualization of the reconstructed model along with measurable accuracy and performance results. The final submission should also include the proposed methodology, system architecture, validation approach, test results, identified limitations and a demonstration using representative medical imaging datasets.

PROBLEM STATEMENT #6 • REF: 6a996c5f239fdeda MEDTECH

Smart Precision Motion Platform for Ophthalmic Imaging

1. Background & Industrial Context:

Modern eye-imaging machines take very detailed pictures of the inside of the eye. To do this, the camera head has to be brought to exactly the right place in front of a seated patient and then held very steady, because even a tiny vibration blurs the image. The patient should not have to be moved or repositioned, and the machine must switch from one eye to the other quickly and safely without ever touching the face. Today this movement is handled by mechanical stages that are either accurate but bulky and expensive, or affordable but not repeatable enough. Operators also need a simple way to make a final fine adjustment by hand, and the machine must know at all times where its own camera head actually is, so that it can stop safely if something jams or slips. Therefore, there is a need for a compact motion platform that is precise, quiet, safe close to a patient's face, easy to build and service, and low in cost.

2. Challenge Statement:

Develop and demonstrate a precision motion platform that positions an eye-imaging head around a seated patient. The platform should move along and around several axes, travel smoothly and repeatably, and settle with very little vibration. It must move between the left and the right eye on its own, along a planned path that always stays outside a defined no-go zone around the patient's face, and it should never require the patient to be repositioned. Position sensors should confirm the actual position at all times, detect slip or lost motion, and trigger a safe stop and recovery if anything goes wrong. Operators should also be able to take over and fine-tune the alignment by hand through an intuitive control, with the speed reducing as the head comes closer to the eye.

3. Expected Deliverables & Outcomes:

Integrated system architecture, motion-envelope model, path-planning logic, closed-loop control strategy, manual alignment interface, costed concept BOM, validation plan, and prototype or simulation evidence.

PROBLEM STATEMENT #7 • REF: 6a996c5e239fdeda MANUFACTURING

Automated Asset Recognition and Intelligent 3D Model Generation from Laser Scan Data

1. Background & Industrial Context:

Industrial plants contain items such as pipes, valves, tanks, pumps, structures, instruments, and cable trays. Laser scanners capture the shape and position of these items as millions of points, creating a three-dimensional picture called a point cloud. This information is useful when old drawings are missing, inaccurate, or no longer match the plant. However, a point cloud is mainly a collection of shapes and does not automatically tell engineers what each item is or how it should be represented in an engineering model. People must inspect the scan, identify objects, choose matching parts from libraries, place them correctly, and check the result. This work takes time and specialist effort, especially when scans contain noise, missing areas, shadows, or crowded equipment. There is a need to reduce this manual work while keeping the final model accurate, traceable, easy to review, and useful for design, maintenance, planning, safety, and future plant changes.

2. Challenge Statement:

Develop and demonstrate a solution that turns laser-scan point-cloud data into engineering information with less manual effort. Participants may choose a plant area, asset types, data sources, tools, AI methods, engineering libraries, and form of POC. The solution should identify selected objects, describe what they are, show where they are located, and create or support an engineering-ready three-dimensional model. Teams may also use drawings, equipment lists, photographs, or information to improve the result. The system should show confidence, highlight uncertain or missing areas, and allow an engineer to review and correct its output. Participants should test the solution using representative scan conditions, including clutter, noise, partial visibility, or incomplete data where appropriate. Results should be compared with a trusted reference using measures such as recognition accuracy, location and size error, model completeness, processing time, and manual effort saved.

3. Expected Deliverables & Outcomes:

1.Point Cloud Asset Recognition Engine:
• AI/ML model to identify objects from point cloud data.
• Automatic classification of: Equipment, Piping, Valves, Structural elements, Instruments, Cable trays, HVAC components
2. Asset Classification Report
• List of identified assets with: Asset Type, Confidence Score, Coordinates/Location, Quantity Count, Export in Excel/CSV format.
3. Intelligent 3D Model Generation
• Automated creation of a 3D engineering model from point cloud data.
• Placement of catalog-based components from engineering libraries.
4. Automated Component Placement
• Rule-based or AI-driven placement of: Pumps, Vessels, Pipes, Supports, Valves & Instruments

PROBLEM STATEMENT #8 • REF: 6a996c5e239fdeda MANUFACTURING

Enterprise Digital Twin of Factory

1. Background & Industrial Context:

Factories generate information about equipment, materials, production, quality, maintenance, energy, safety, schedules, and cost, but it is often fragmented across machines, systems, documents, and people. This makes it difficult to understand operations, anticipate the effect of changes, and evaluate improvement opportunities before acting on the physical plant. Existing digital representations may focus on isolated assets or dashboards without maintaining a dependable relationship between actual operations and their virtual representation. Differences in data availability, timing, accuracy, asset definitions, and process context further limit trust and reuse. A factory digital twin should provide meaningful operational insight rather than only a visual model, but suitable scope, architecture, technologies, and fidelity may vary by use case. The challenge is to discover a practical way to represent relevant factory behavior, connect physical and virtual information, explore alternatives, and demonstrate measurable value over time reliably while remaining scalable, secure, maintainable, and realistic for existing manufacturing environments.

2. Challenge Statement:

Design and demonstrate an innovative proof of concept for a factory digital twin that addresses a defined manufacturing need. Teams may select an appropriate process, production area, asset group, material flow, utility system, or operational decision as the focus. The solution should establish a meaningful relationship between the factory context and its virtual representation, keep the twin current and clearly demonstrate how users gain insight or evaluate alternatives. Teams may choose the modeling approach, data sources, sensing methods, hardware, software, analytics, visualization, simulation, artificial intelligence, and interaction model. The POC should explain its assumptions, boundaries, data requirements, intended users, and measurable success criteria. It should show how the concept could evolve beyond the demonstration without claiming plant-wide completeness. The emphasis is on originality, usefulness, reasoning, and evidence that the proposed twin solves the selected problem.

3. Expected Deliverables & Outcomes:

- Clearly defined manufacturing problem, users, scope, assumptions, and success criteria.
- Working proof of concept demonstrating the proposed physical-to-virtual relationship.
- Suitable representation of the selected factory process, assets, behavior, or decisions.
- Evidence that the twin remains current using the team's chosen data or interaction method.
- Demonstration of operational insight, prediction, simulation, optimization, or another justified value.
- Evaluation against a relevant baseline, scenario, or reference case.
- Scalability, safety, security, maintainability, and integration considerations.

PROBLEM STATEMENT #9 • REF: 6a996c5f239fdeda MANUFACTURING

Spatial-Analysis-to-BOM Generator for LightingDesign

1. Background & Industrial Context:

Specifying lighting and control products for a room requires manual measurement, layout interpretation, and product selection, then hand-building a BOM. This is slow, inconsistent, and error-prone, and weak fitment leads to rework and returns.

2. Challenge Statement:

Build a prototype that analyzes a room's spatial layout (image/scan/floorplan), detects dimensions, fixture positions, and control zones, and automatically generates a validated lighting/controls BOM from the Nexus catalog. Computer vision extracts spatial features; a rules-plus-LLM layer selects compatible products, checks coverage, and explains each selection.

3. Expected Deliverables & Outcomes:

A 'room-to-BOM' prototype that ingests a room scan/floorplan, derives the layout, and outputs a validated, explained Nexus product BOM.

PROBLEM STATEMENT #10 • REF: 6a996c5e239fdeda MANUFACTURING

AI Agents for Resilient Supply Chain

1. Background & Industrial Context:

Supply-chain planning spans demand sensing, inventory, supplier risk, and logistics across disconnected systems. Disruptions are detected late and responses are manual and siloed, causing stockouts, excess inventory, and missed commitments.

2. Challenge Statement:

Develop a multi-agent supply-chain POC for a defined three-echelon network facing one disruption: a seven-day shutdown of a critical supplier. Use a supplied or reproducible dataset containing demand, inventory, capacity, lead time, transport cost, and service constraints. Demand, inventory, supplier-risk, and logistics agents may propose actions, but a coordinator must reconcile them using explicit business constraints and a mathematical optimization baseline. The system shall recommend expediting, alternate sourcing, allocation, or rescheduling; explain trade-offs; and require planner approval. Compare the agent solution with reorder rules and optimization-only planning over identical scenarios.

3. Expected Deliverables & Outcomes:

1. Reproducible network dataset, disruption scenarios, constraints, and cost assumptions.
2. Specialized agents with defined inputs, outputs, permissions, and shared schema.
3. Coordinator with conflict resolution, constraint validation, and planner approval.
4. Reorder-rule baseline and mathematical optimization baseline.
5. What-if interface showing recommendation, service/cost trade-offs, confidence, and rationale.
6. Evaluation using fill rate, backorders, total landed cost, recovery time, inventory days, constraint violations, runtime, and recommendation stability.
7. Failure, data-quality, and governance analysis.

PROBLEM STATEMENT #11 • REF: 6a996c5e239fdeda MANUFACTURING

AI-Driven Engineering Change Management for Resilient Material and Sourcing Decisions

1. Background & Industrial Context:

Engineering changes can alter part numbers, bills of material, specifications, suppliers, effective dates, and production requirements. The change record may be managed in an engineering system, while inventory, purchase orders, sourcing restrictions, lead times, and demand are maintained in material-management applications. When sources are not connected, planners identify affected materials and decide when old parts should be consumed, cancelled, or written off and when replacements should be ordered. Delayed analysis can create obsolete inventory, shortages, duplicate purchases, production disruption, and late implementation. Phase-in and phase-out decisions are difficult when open orders, existing stock, supplier constraints, restricted purchasing groups, or multiple plants are involved. Teams may lack a consistent view of ownership, approvals, risks, and completion status across the change lifecycle. The core need is a traceable way to connect an approved engineering change with material and sourcing consequences so planners can identify exposure early and coordinate timely actions before implementation.

2. Challenge Statement:

Develop an AI-assisted proof of concept for material-impact assessment and phase-in/phase-out planning for one defined engineering-change scenario. Using synthetic, sanitized, data, the POC should ingest a change record, affected bill-of-material items, effective date, inventory balances, open purchase orders, demand, lead times, supplier information, and restricted-purchasing indicators from simulated ECM and SAP-like sources. It should identify impacted old and new materials, calculate projected inventory exposure, detect likely shortages or excess stock, and highlight sourcing constraints. A rules-based workflow should recommend actions such as consume existing stock, revise an effective date, expedite or defer an order, seek an alternate source, or request planner review. AI may summarize risks, prioritize exceptions, and explain recommendations. The dashboard should show change status, material impact, owners, decisions, and an auditable action log.

3. Expected Deliverables & Outcomes:

- Defined engineering-change scenario, data model, assumptions, and workflow.
- Simulated ECM and SAP-like datasets for BOM, inventory, demand, purchase orders, suppliers, and restrictions.
- Material-impact engine identifying old and new parts, exposure, shortages, excess, and sourcing constraints.
- Phase-in/phase-out recommendation rules with explainable risk prioritization.
- Planner dashboard with change status, material exceptions, owners, decisions, and action log.

PROBLEM STATEMENT #12 • REF: 6a996c5e239fdeda MANUFACTURING

Digital Path Teaching and Localization System for Variable Manufacturing Environments

1. Background & Industrial Context:

Industrial automation systems are traditionally designed to execute highly repeatable tasks within controlled operating environments. In applications such as adhesive dispensing, inspection, machining, and material handling, equipment often follows a predefined path programmed using fixed coordinates. While such approaches work effectively under ideal conditions, real-world manufacturing environments frequently experience variations in part positioning, dimensional tolerances, fixture accuracy, and product geometry.

These variations can lead to deviations between the programmed path and the actual workpiece, resulting in reduced process quality, increased rework, and lower operational efficiency. Adapting automation systems to accommodate such variations often requires manual intervention, reprogramming, or dedicated fixtures, which can increase setup times and operational costs.

There is a growing need for intuitive and flexible solutions that enable operators to rapidly capture spatial information from physical workpieces and communicate this information directly to industrial equipment. Such systems can empower adaptive automation by allowing manufacturing processes to dynamically adjust to real-world variations while minimizing engineering effort, setup time, and production downtime.

2. Challenge Statement:

Develop a wireless digital pen capable of recording the spatial coordinates of user-selected points on a planar work surface through a button-triggered input. The recorded coordinates shall be communicated to an XY positioning stage, which will automatically move to each stored location for point tracing or verification.

3. Expected Deliverables & Outcomes:

Take a laser marked grid plate such that the grid locations are known. Now draw a random curve on the grid plate. Use the wireless digital pen to teach the coordinates of the points where the curve and the grid line meet.

Transmit these coordinates to the XY positioning system.

The positioning system shall have a laser pointer attached pointing to the grid plate. Upon reaching each point, mark the tracked point with a marker. The difference between these points must be 300 µm or better, measured as the deviation between the point recorded by the digital pen and the corresponding position reached by the XY stage. The solution should provide reliable wireless communication, repeatable coordinate acquisition, and accurate point-to-point positioning.

The report should include detailed explanations, diagrams, and calculations to showcase the design

PROBLEM STATEMENT #13 • REF: 6a996c5f239fdeda MANUFACTURING

Smart Digital Setup Assistance for Rapid and Error-Free Assembly Line Changeovers

1. Background & Industrial Context:

Product changeovers on manual assembly lines often depend heavily on the experience, memory, and working methods of individual operators. Experienced personnel may restore production quickly, while others require more time or encounter difficulties that affect first-piece quality, scrap, rework, capacity, and resource consumption. Setup knowledge may be distributed across paper procedures, digital files, informal notes, and undocumented practice, making it difficult to keep instructions current or transfer expertise when people change roles. Supervisors may have limited visibility into progress, delays, repeated actions, exceptions, or differences between shifts, products, and workstations. A digital solution could make changeovers more consistent and provide evidence for improvement, but a rigid workflow may overlook legitimate variation, operator insight, accessibility needs, or unexpected shop-floor conditions. The challenge therefore extends beyond digitizing an existing procedure. It requires teams to consider how knowledge is captured, how guidance adapts to context, how actions are confirmed without creating unnecessary burden, and how information supports learning rather than surveillance. Any approach must remain usable in the realities of a shop floor, including intermittent connectivity, changing products, mixed equipment, safety requirements, data governance, and the need to integrate with existing work practices and systems.

2. Challenge Statement:

Design and demonstrate an innovative proof of concept that makes manual assembly-line changeovers or equipment setup faster, more consistent, easier to learn, and less dependent on individual experience. Participants may select a representative process and define the users, operating context, form of guidance, information sources, interaction method, level of automation, and implementation platform. The solution should help operators access the right knowledge at the right time, recognize critical or missed activities where appropriate, and capture useful evidence about progress, delays, exceptions, quality, or learning needs. Teams are encouraged to question the current process rather than merely reproduce paper instructions on a screen. Approaches may use software, connected tools, physical interfaces, sensing, visualization, artificial intelligence, workflow redesign, or other justified combinations. The POC should accommodate changing products and procedures, intermittent connectivity, legitimate deviations, and different operator needs while maintaining safety, security, revision control, and respectful use of workforce data. Participants should define meaningful baseline and outcome measures and demonstrate how the concept could exchange information with relevant shop-floor systems. The goal is a practical and scalable improvement concept, not a prescribed device, workflow, or technology stack.

3. Expected Deliverables & Outcomes:

- Clearly defined changeover problem, users, baseline, assumptions, and success measures.
- Working POC using participant-selected software, hardware, workflow, or hybrid methods.
- Context-appropriate guidance and support for critical activities, exceptions, and learning.
- Evidence capture and insights that help improve time, quality, consistency, or resource use.
- Demonstration on a representative setup with measurable comparison to the baseline.

PROBLEM STATEMENT #14 • REF: 6a996c5f239fdeda MANUFACTURING

Transforming Legacy Factories into Intelligent Manufacturing Facilities

1. Background & Industrial Context:

Legacy manufacturing plants often contain ageing machines, standalone controls, manual records, disconnected quality systems, and limited visibility. Production, maintenance, energy, safety, and material-flow information may be captured in separate spreadsheets or applications, preventing teams from seeing how losses are connected. Equipment may not support modern communication protocols, and replacing every asset is rarely affordable or practical. As a result, modernization decisions can become technology-led rather than problem-led, with unclear priorities, uncertain returns, and disruption risks. Plant teams need to identify where downtime, bottlenecks, quality losses, excess energy use, unsafe conditions, or manual effort have the greatest impact before selecting automation or digital solutions. They must also consider existing infrastructure, cybersecurity, workforce skills, maintainability, and phased implementation. The core need is a repeatable method for assessing one production area, establishing performance baselines, identifying feasible improvement opportunities, and creating a justified roadmap that balances operational value, cost, risk, compatibility, and implementation effort.

2. Challenge Statement:

Develop a smart-manufacturing modernization proof of concept for one defined production line, cell, or utility area using synthetic, historical, or mentor-approved plant data. The POC should document the current process, equipment, data sources, material flow, major losses, and operational constraints. It should define baseline measures for selected outcomes such as availability, cycle time, throughput, quality, energy, or safety. Teams should choose one or two priority use cases, for example condition monitoring, downtime classification, energy monitoring, production visibility, or bottleneck detection, and demonstrate them using simulated or low-cost sensor data. A dashboard should present current status, trends, exceptions, and the calculated business effect of identified losses. The solution should produce a future-state architecture and prioritized roadmap showing required sensors, connectivity, data flow, cybersecurity controls, integration points, estimated effort, benefits, dependencies, and risks.

3. Expected Deliverables & Outcomes:

- Defined production area, process map, asset list, constraints, and modernization objectives.
- Current-state digital-maturity and loss assessment with baseline performance measures.
- Prepared dataset or sensor simulation for one or two selected use cases.
- Working monitoring or analytics POC with alerts and trend visualization.
- Dashboard showing status, losses, operational impact, and improvement opportunities.
- Future-state architecture covering sensors, connectivity, data flow, integration, and cybersecurity.
- Prioritized roadmap with benefits, effort, dependencies, risks, and indicative ROI assumptions.

PROBLEM STATEMENT #15 • REF: 6a996c5f239fdeda MEDTECH

Aqua Birth - Smart Water Birth System with Underwater Wireless Communication for Maternal & Fetal Monitoring

1. Background & Industrial Context:

More and more hospitals now offer water birth, where the mother spends part of her labour in a warm pool of water. Being in water helps her relax, reduces pain and supports a more natural delivery. But it creates a practical problem for the medical team. The equipment normally used to watch the health of the mother and the baby is wired, and wires cannot be used safely in water. Water also blocks ordinary wireless signals such as Bluetooth and Wi-Fi, so a normal wireless sensor placed under water quickly loses its connection. As a result, doctors and midwives often have to ask the mother to sit up or step out of the pool from time to time so that checks can be done by hand. This interrupts her comfort and, more importantly, leaves gaps in which an early warning sign in the mother or the baby could be missed. Hospitals therefore need a safe, waterproof and cable-free way to keep watching both mother and baby continuously while she stays in the water.

2. Challenge Statement:

Design and build a working demonstration of a waterproof, cable-free monitoring system for water birth. Small wearable or floating sensors on the mother should keep measuring her heart rate, blood pressure, body temperature, blood oxygen level, and the timing and strength of her contractions. A second sensor should follow the baby's heart rate, movement and any early sign of distress. All this data must travel through the water to a bedside screen without wires, and appear almost immediately so that staff can act in time. Teams will have to find a way of sending signals through water. The system must be fully sealed, run for the length of a long labour on a small battery, stay comfortable while the mother moves, and raise a clear alarm whenever a reading becomes unsafe.

3. Expected Deliverables & Outcomes:

Underwater communication module (acoustic, ultrasonic, magnetic induction, optical, or hybrid)
Wearable maternal sensors
Fetal monitoring sensors, interface
Edge AI analytics
Mobile/Cloud dashboard
Emergency alert generation

PROBLEM STATEMENT #16 • REF: 6a996c5e239fdeda ENERGY

AI Reasoning model for an efficient commercial refrigeration system

1. Background & Industrial Context:

Commercial refrigerators, freezers, and chillers are used in restaurants, warehouses, hospitals, and businesses to keep products at safe temperatures. Their cooling needs change when doors are opened, items are added, outside temperature changes, or frost builds up inside the system. Many units follow fixed settings and timed defrost cycles, even when conditions are different. This can waste electricity, cause uneven cooling, increase wear on the compressor, or allow stored products to become too warm. Problems such as weak power supply, unusual vibration, blocked airflow, failing parts, or poor maintenance may also go unnoticed until the equipment stops working. Operators often have limited information about why energy use has increased or when a fault is developing. There is a need for a practical, reliable way to understand changing cooling demand, reduce unnecessary energy use, detect problems early, and maintain safe storage conditions without making the system difficult or expensive to operate.

2. Challenge Statement:

Develop and demonstrate a system that improves the efficiency and reliability of a commercial refrigerator, freezer, or chiller. Participants may choose the equipment, users, operating conditions, data sources, hardware, software, and technical approach. The solution should understand changing cooling needs and help decide when actions such as cooling or defrosting may be required. It may also identify unusual behaviour linked to the compressor, power supply, airflow, vibration, doors, temperature, or other conditions. Teams should explain how the system handles missing or incorrect data, changing loads, sensor problems, and situations where it is uncertain. The POC should compare its performance with the normal operating method using measures such as energy use, temperature stability, defrost effectiveness, warning time, false alarms, and equipment health.

3. Expected Deliverables & Outcomes:

The problems can be addressed by established simulations followed by demonstrating the innovations in a live commercial refrigerator such as deep freezer or a chiller

PROBLEM STATEMENT #17 • REF: 6a996c5e239fdeda MANUFACTURING

AI Tool for Serviceability Evaluation- Via Design Data

1. Background & Industrial Context:

Serviceability is a critical product-design attribute because it influences maintenance time, repair cost, product availability, technician safety, customer satisfaction, and total cost of ownership. In appliances and other complex products, engineers routinely evaluate manufacturability and assembly, but serviceability is often reviewed later and through manual expert judgement. The assessment can vary between reviewers and may not consistently account for component accessibility, disassembly sequence, tool clearance, fastener type, replacement effort, ergonomics, modularity, or service-related hazards. As a result, design weaknesses may remain hidden until a physical prototype is built or field failures occur, when modifications are slower and more expensive. Existing CAD models already contain geometric and product-structure information that could support an earlier, repeatable assessment, while historical repair records can provide evidence about actual service effort. A standardized AI-assisted approach is therefore needed to analyze design data, explain the factors affecting serviceability, compare design alternatives, and provide actionable recommendations before design release. The solution should support engineering judgement rather than replace formal design reviews.

2. Challenge Statement:

Develop an AI-assisted proof of concept that imports a selected CAD or 3D product assembly, identifies serviceable components and their relationships, and generates an explainable serviceability score using configurable engineering criteria. The POC should evaluate access paths, disassembly steps, tool and hand clearance, fastener complexity, component replaceability, estimated repair time, ergonomics, modularity, and safety constraints. It should highlight obstructing parts on the 3D model, show how each criterion contributes to the overall and subsystem scores, compare design revisions, and recommend practical improvements. The system should preserve traceability to the source geometry, bill of materials, rules, and any historical service data used. Validation should use approved designs or controlled test assemblies and compare the generated assessment with expert review. A hardware-assisted validation module may use a depth camera, AR headset, or instrumented physical mock-up to capture real access, reach, visibility, and tool-clearance evidence and compare it with the CAD-based prediction.

3. Expected Deliverables & Outcomes:

- CAD/3D assembly importer and serviceable-component identification.
- Configurable scoring model with criterion-level explanations.
- Accessibility, disassembly, tool-clearance, replaceability, ergonomics, modularity, repair-time, and safety assessment.
- 3D heat map showing constraints and recommended design improvements.
- Design-revision comparison dashboard with traceable evidence.
- Validation against expert assessments and agreed accuracy measures.

PROBLEM STATEMENT #18 • REF: 6a996c5f239fdeda MANUFACTURING

Smart Visual Planning Tool for Lift Modernization

1. Background & Industrial Context:

Many lifts in older buildings need to be updated to improve safety, reliability, energy use, accessibility, and appearance. Before starting such work, building owners usually want to understand what can be changed, how the upgraded lift may look, which parts are suitable, and what the work may involve. Today, this process often depends on site visits, photographs, measurements, product catalogues, and proposals prepared manually by sales and engineering teams. Preparing different options can take time, and customers may find technical drawings or written descriptions difficult to understand. It is also possible for a suggested part or design to be unsuitable for the available space or existing lift arrangement. A simple digital tool could make the planning process clearer and faster. By using site photographs, videos, basic measurements, and an approved list of modernization options, the tool could show realistic before-and-after views, explain the proposed changes, and prepare an initial proposal for review. The final decision must remain with qualified professionals.

2. Challenge Statement:

Design and demonstrate a proof of concept that helps users plan and visualize the modernization of an existing lift. The system should accept photographs or short videos of a lift cabin, entrance, or lobby together with basic site measurements and user preferences. It should identify the main visible areas, such as doors, panels, lighting, controls, walls, flooring, and display units, and suggest suitable upgrade options from a provided product list. The tool should create at least two realistic before-and-after visualizations while keeping the size, viewpoint, surrounding structure, and unchanged parts consistent with the original scene. It should clearly show which items are proposed, why they were selected, and whether any important information is missing. The system should also prepare a simple draft proposal containing the selected options, assumptions, reference images, and approximate work scope. Users must be able to edit or reject suggestions, and a qualified person must review all results before they are shared with a customer.

3. Expected Deliverables & Outcomes:

1. Bounded lift catalog, three scene datasets, site-dimension template, and gold annotations.
2. Scene calibration, component recognition, and geometry-fit module with declared tolerance.
3. Two catalog-valid modernization configurations with rule checks.
4. Photorealistic before/after renderings preserving geometry and nonmodified elements.
5. Citation-grounded proposal/brochure with assumptions, unknowns, and reviewer approval.
6. Evaluation using fit error, component-selection accuracy, rule violations, perceptual realism study, factual error rate, and time saved versus manual creation.
7. Failure cases and deployment constraints.

PROBLEM STATEMENT #19 • REF: 6a996c5f239fdeda MEDTECH

AI-Powered Leadless Cardiac Monitoring for Early Arrhythmia Detection

1. Background & Industrial Context:

Intermittent heart-rhythm abnormalities may occur outside a clinic and can be difficult to capture during a short examination. Conventional electrocardiogram systems provide valuable information, but their electrodes, leads, placement requirements, and clinical workflow may limit comfortable monitoring over extended periods. Consumer wearables collect signals related to pulse, movement, oxygen saturation, respiration, and other physiological activity, yet these measurements can be noisy, incomplete, affected by motion, and insufficient for dependable rhythm assessment on their own. A practical monitoring approach must balance signal quality, comfort, battery life, privacy, usability, and the need to distinguish meaningful events from artifacts. Available datasets may also contain class imbalance, inconsistent labels, different devices, and limited examples of rare conditions. The challenge is to explore how a minimally intrusive system could capture or combine useful signals, recognize possible rhythm abnormalities, communicate uncertainty clearly, and support timely review without creating false reassurance, unnecessary alarm, or unsupported diagnostic claims.

2. Challenge Statement:

Design and demonstrate a proof of concept for continuous or repeated cardiac-rhythm monitoring outside a conventional multi-lead clinical setup. Teams may define the target users, monitoring context, rhythm events, sensing approach, device form, data sources, and interaction model. The solution may use electrical, optical, mechanical, acoustic, motion, or contextual signals, together with signal processing, artificial intelligence, or other justified methods. It should address signal quality, artifacts, uncertainty, personalization, and the difference between screening information and clinical diagnosis. Participants should show how potential abnormalities are detected or prioritized, how evidence is presented for review, and how the user is informed when data quality is insufficient.

3. Expected Deliverables & Outcomes:

- Clearly defined monitoring need, users, target events, assumptions, and success criteria.
- Representative dataset, simulation, or safe nonclinical sensing demonstrator chosen by the team.
- Working POC for signal acquisition or ingestion, quality assessment, and event detection or prioritization.
- Method for handling artifacts, missing data, uncertainty, and individual variation.
- User or reviewer interface showing evidence, confidence, alerts, and insufficient-quality conditions.
- Evaluation against expert labels, public benchmarks, or another justified reference.
- Safety, privacy, usability, power, and scalability considerations.

PROBLEM STATEMENT #20 • REF: 6a996c5f239fdeda MEDTECH

Non-Invasive Knee Movement Tracking for Gait Assessment

1. Background & Industrial Context:

The human knee is not a simple hinge. It moves with six degrees of freedom: flexion-extension, internal-external rotation, abduction-adduction, and anterior-posterior, medial-lateral and superior-inferior translation. Clinical assessment today captures only a fraction of this. Goniometers and standard gait labs report flexion angle reliably but characterise rotation and translation poorly, and marker-based optical systems suffer from soft-tissue artefact at the thigh and shank. As a result, clinicians assessing ACL deficiency, meniscal injury, osteoarthritis progression or post-operative recovery work with an incomplete picture of joint behaviour, particularly during dynamic tasks such as stair descent, pivoting and squatting where instability actually presents. Rehabilitation protocols and implant design decisions are therefore based largely on sagittal-plane data. A low-cost, non-invasive system that resolves all six degrees of freedom during functional movement would give clinicians objective, repeatable evidence of joint mechanics, support earlier detection of instability, and allow rehabilitation progress to be measured rather than estimated.

2. Challenge Statement:

Design and validate a non-invasive system that measures six-degree-of-freedom knee kinematics during functional movement, and present the output in a form a clinician can interpret. Teams may combine wearable inertial measurement units, depth or multi-camera vision, flexible strain sensing, or other non-invasive modalities, together with a sensor-fusion algorithm that resolves rotation and translation simultaneously. The system must operate outside a dedicated gait laboratory, tolerate soft-tissue movement, and remain usable by a single operator. Teams are expected to define a movement protocol (for example gait, sit-to-stand, stair descent and a controlled pivot), acquire data from healthy volunteers, and validate their output against an accepted reference such as an optical motion-capture system or an instrumented rig. Accuracy targets: rotational error within 5 degrees and translational error within 5 mm root-mean-square against the reference, sustained across the full movement cycle. Implantable or in-vivo sensing is explicitly out of scope; ligament-level strain measurement may be discussed only as future work.

3. Expected Deliverables & Outcomes:

The developed system to demonstrate the ability to directly measure skeletal kinematics in 6-DOF despite soft tissue deformations between the transducer and the bony surface.

PROBLEM STATEMENT #21 • REF: 6a996c5e239fdeda ENERGY

Reimagining Ceiling Airflow Designing a Safe Bladeless Ceiling Fan

1. Background & Industrial Context:

Ceiling fans are widely used because they are simple and affordable, but their exposed blades can create safety concerns in low rooms, schools, hospitals, offices, and spaces used by children. Blades collect dust and may be difficult to clean. An enclosed or bladeless-looking fan could improve safety and appearance, but hiding the moving parts creates new problems. Air must travel through internal passages and outlets, which can reduce airflow, create uneven cooling, increase noise, and use more electricity. The unit may also become bulky, costly, difficult to repair, or harder to install on a ceiling. Existing bladeless fans show that the basic idea is possible, so a new design must offer a clear advantage for overhead use. There is a need to explore whether a safe enclosed ceiling-air system can provide useful comfort while balancing airflow, noise, energy use, size, weight, cost, maintenance, durability, cleanliness, installation, and ease of manufacture.

2. Challenge Statement:

Develop and demonstrate a working physical POC for a ceiling-mounted air-moving system without exposed rotating blades. Participants may choose the shape, airflow principle, internal mechanism, materials, controls, manufacturing method, and appearance. The prototype should create useful airflow in a chosen room or test area and show how the design improves safety or another important user need. Teams should compare it with a suitable conventional fan or reference using measures such as air movement, comfort coverage, noise, power use, size, weight, cost, vibration, cleaning, maintenance, and installation. Calculations, airflow simulation, or digital models may support the design but cannot replace the physical prototype. The solution should include protection against contact with moving parts, overheating, electrical faults, loose mounting, and other relevant risks.

3. Expected Deliverables & Outcomes:

Product requirement document.
Benchmarking report.
Concept architecture.
Industrial design renders.
3D CAD packaging.
Airflow path design.
CFD report.
Structural CAE report.
Motor and impeller selection report.
DFM/DFA report.
Prototype drawings.
Alpha prototype.
Test report.
Design refinement report.
Final CAD, BOM, drawings, and assembly instructions.

PROBLEM STATEMENT #22 • REF: 6a996c5f239fdeda MEDTECH

Self Calibrating and Healing Medical Devices Using AI and Digital Twins

1. Background & Industrial Context:

Hospitals run on measuring equipment: patient monitors, infusion pumps, ventilators, ECG machines, scanners and laboratory analysers. Every one of these has to be checked and adjusted from time to time, because sensors slowly drift away from their true reading. Today that check is done by hand. A service engineer visits the site, takes the device out of use, compares it against a reference and corrects it. This is slow and expensive, it depends on skilled people being available, and it is itself open to human error. Worse, drift does not wait for the service calendar. A device can quietly start reading a little high or a little low between two visits, and a doctor may take a decision based on a wrong number without knowing it. The industry now wants equipment that can watch its own accuracy, notice when it is going out of tune, warn before it fails, and correct itself wherever possible.

2. Challenge Statement:

Build a working demonstration of a medical device that can look after its own calibration. Pick one or two device types and their sensors, for example pressure, flow, temperature, optical or ECG, and create a system that continuously checks whether the readings can still be trusted. The solution should spot slow drift early, tell real drift apart from ordinary noise and from changes in room temperature or humidity, and estimate how long a sensor has left before it must be corrected or replaced. Where the design allows, the device should correct itself automatically; where it cannot, it should clearly assit on what to do and when. A digital twin, that is a software copy of how the device is expected to behave, can be used to compare expected against actual behaviour.

3. Expected Deliverables & Outcomes:

Expected Solution :
Intelligent Sensor Layer
Embedded sensor diagnostics
Sensor redundancy
Self-test routines
Digital Twin Layer
Create a virtual replica of:
Device behavior
Sensor performance
Expected operating parameters
AI Analytics Engine :
Drift prediction
Anomaly detection
Root-cause analysis
Calibration optimization
Clinical Dashboard :
Device status
Calibration confidence score
Risk alerts
Predictive maintenance recommendations

PROBLEM STATEMENT #23 • REF: 6a996c5f239fdeda MOBILITY

Validating Trust in the Software-Defined Vehicle

1. Background & Industrial Context:

A modern car is now mostly software. Features arrive and get updated over the air. That is convenient, but a bug in a car can mean the brakes assist at the wrong moment, the car drifts out of its lane, or the battery is managed unsafely -- and it happens across thousands of vehicles at once, because they all received the same update.

The real problem is testing. The traditional way of checking car software involved months of manual test cycles and expensive lab rigs with real hardware wired in. That was fine when a car's software changed once a year. It does not work when engineers want to ship an update every week.

So the industry is shifting towards testing with simulation and AI instead of physical rigs -- running the car through thousands of virtual scenarios overnight rather than driving it around a test track for months. The technology largely exists. What is missing is confidence: how do you prove that a fast, automated test process has genuinely caught the dangerous bugs before the update reaches real drivers?

2. Challenge Statement:

Propose and demonstrate a testing or validation approach -- using simulation, virtual hardware-in-the-loop, or AI-driven test generation -- that detects a software regression or safety-critical fault in a vehicle feature (e.g., ADAS, battery management, or an infotainment-to-safety interaction) before deployment. The approach must be usable within a compressed OTA release cycle measured in days, and teams must clearly articulate how test cases are generated, prioritised, and assessed for coverage. Teams may choose to work with vehicle simulators (e.g., CARLA), synthetic sensor data, or mocked ECU environments.

3. Expected Deliverables & Outcomes:

A proof-of-concept that detects at least one deliberately injected fault in a simulated vehicle environment, delivered as a runnable demo. Teams must state explicit limitations (e.g. fault classes not caught) and explain how the approach would integrate into a real OTA release pipeline.

PROBLEM STATEMENT #24 • REF: 6a996c5f239fdeda MEDTECH

AI-Powered Multimodal Surgical Assistant for Real-Time Operating Room Guidance

1. Background & Industrial Context:

Critical and Complicated surgery are time consuming and at time very much dependent on limited number of specialists. There is need for guided context decision support from experts to bring in scale.

2. Challenge Statement:

Develop a nonclinical conversational assistant for one simulated operating-theatre workflow: the pre-incision safety checklist. Using scripted scenarios, synthetic patient data, and one modality--full-duplex speech--the assistant shall recognize checklist items, retrieve approved protocol content, track completion, identify omissions or contradictory responses, and provide concise prompts. It must handle interruptions, overlapping speech, noise, and uncertain recognition; never make diagnosis or autonomous treatment decisions; and require the designated clinician to confirm every checklist completion. Compare with a static digital checklist in controlled simulations.

3. Expected Deliverables & Outcomes:

1. Approved checklist knowledge base, synthetic cases, scripted omissions, and audio-noise scenarios.
2. Streaming speech recognition and speaker/turn handling for the bounded workflow.
3. Protocol-grounded dialogue manager with state tracking, confidence, and abstention.
4. Human confirmation, override, correction, and complete timestamped audit trail.
5. Simulation interface clearly labeled nonclinical.
6. Evaluation versus static checklist using item-completion accuracy, omission-detection recall, false prompts, response latency, interruption recovery, task time, and user workload.
7. Safety case, privacy analysis, and clinical-validation roadmap.

PROBLEM STATEMENT #25 • REF: 6a996c5f239fdeda MOBILITY

AI-Enabled Suspension Digital Twin with Sensor Fusion for Ride Comfort Prediction

1. Background & Industrial Context:

Ride-comfort development requires engineers to understand how road disturbances, vehicle speed, loading, suspension behavior, and seat vibration affect occupants. Simulation and physical testing provide evidence, but simulation models may not match the tested vehicle, subjective ratings vary, and proving-ground programs cannot cover every operating condition. Onboard sensors provide incomplete and noisy views of body motion, wheel behavior, suspension travel, and road inputs. Differences in sampling rate, delay, bias, and temporary signal loss make direct interpretation difficult. Treating comfort prediction and sensor estimation as separate activities creates duplicated models, datasets, dashboards, and validation effort. A unified approach is needed in which synchronized sensor data estimates road and suspension states, a physics-based digital twin predicts vehicle response, and an AI model converts these responses into a repeatable comfort indicator. The framework must quantify uncertainty, preserve physical behavior, and clearly distinguish validated operation from unsupported conditions before formal engineering decisions are made safely.

2. Challenge Statement:

Develop an AI-assisted suspension digital-twin proof of concept that combines sensor fusion, road-state estimation, vehicle-response simulation, and ride-comfort prediction for a quarter-car or half-car model. Using simulated, public data, the POC should ingest body and wheel acceleration, suspension travel, wheel speed, vehicle speed, and inertial measurements. It should synchronize signals, filter noise, detect dropouts, and estimate selected road and suspension states using a justified fusion method. These estimates should drive a physics-based model and an AI/ML model that predicts an objective comfort score using an agreed vibration method. The dashboard should compare raw signals, fused states, simulated response, predicted comfort, confidence, and residual error across defined road, speed, load, and damping scenarios. For a semi-active case, the system may compare predefined damping settings and recommend the best evaluated option.

3. Expected Deliverables & Outcomes:

- Defined sensor set, scenarios, assumptions, comfort metric, and system architecture.
- Synchronized dataset with noise, delay, dropout, and anomaly handling.
- Sensor-fusion model estimating road and suspension states with uncertainty.
- Quarter-car or half-car digital twin with configurable road, load, and damping inputs.
- AI/ML ride-comfort prediction model with confidence and residual error.
- Dashboard integrating raw signals, fused estimates, simulation, comfort score, and validation results.

PROBLEM STATEMENT #26 • REF: 6a996c5e239fdeda ENERGY

AI-Driven Closed-Loop Control Optimization

1. Background & Industrial Context:

The problem: Tuning control parameters for advanced DCDC and AC-DC topologies is a highly iterative, manual process... especially when trying to maintain stability across varying loads and transient conditions...
the challenge: develop an AI/ML model to dynamically auto-tune or optimize closed-loop control parameters... the goal is to achieve optimal transient response, minimize overshoot, and ensure robust stability without exhaustive manual trial and error...

2. Challenge Statement:

Develop an AI-driven closed-loop tuning POC for one simulated plant: a bidirectional EV fast-charger DC-link voltage controller under step-load and grid-voltage disturbances. Use a provided or reproducible MATLAB/Simulink or Python plant model with fixed voltage, current, stability, and switching constraints. The learning or optimization method shall tune PI gains online within certified bounds, fall back to a safe baseline controller when confidence or constraints fail, and log every gain change. Compare against a manually tuned PI controller over identical operating points, parameter variations, sensor noise, and disturbances. No connection to live high-voltage hardware is required.

3. Expected Deliverables & Outcomes:

1. Reproducible charger plant model, disturbance suite, constraints, and manual PI baseline.
2. Online gain-tuning algorithm with bounded action space and stability/safety monitor.
3. Safe fallback, rollback, and complete tuning audit log.
4. Simulation dashboard for voltage, current, gains, constraints, and controller state.
5. Evaluation using settling time, overshoot, integral absolute error, constraint violations, stability margin, computation time, and robustness to model variation/noise.
6. Ablation study, failure analysis, and controlled hardware-in-loop roadmap.

PROBLEM STATEMENT #27 • REF: 6a996c5e239fdeda MANUFACTURING

AI-for-Design Generative Engineering Assistant

1. Background & Industrial Context:

Electrical and panelDesign involves repetitive selection, sizing, and compliance tasks. Engineers spend time on routine configuration and standards checks rather than value-addDesign, and standards expertise is unevenly distributed across teams.

2. Challenge Statement:

Develop a generative engineering assistant for one bounded product family: low-voltage motor-starter panels using a supplied component catalog and a versioned subset of IEC/UL design rules. The assistant shall convert a structured natural-language design brief into candidate component selections, sizing calculations, a simple panel layout, and an explained compliance report. All numeric sizing and rule checks must be executed by deterministic calculators or a rules engine; the language model may interpret intent and explain results but may not invent components, ratings, or standards. Ambiguous or missing requirements must trigger questions, and an engineer must approve the final draft.

3. Expected Deliverables & Outcomes:

1. Versioned product catalog, rule set, test briefs, and engineer-approved gold designs.
2. Brief parser that identifies requirements, ambiguity, and missing inputs.
3. Deterministic sizing and compliance engine with traceable calculations.
4. Catalog-constrained component selection and simple panel-layout generation.
5. Explained design package with citations, assumptions, violations, and approval workflow.
6. Evaluation using requirement-extraction accuracy, component accuracy, compliance precision/recall, sizing error, hallucination rate, reviewer corrections, and time saved.
7. Rule-update and failure-case demonstration.

PROBLEM STATEMENT #28 • REF: 6a996c5e239fdeda MANUFACTURING

Object-Detection Model for Battery Assembly & Maintenance

1. Background & Industrial Context:

Battery assembly and maintenance for aerospace systems require correct component orientation and verified assembly steps. Manual visual checks are inconsistent and miss orientation/placement errors, creating safety and rework risk in ITAR-sensitive environments.

2. Challenge Statement:

Train an object-detection/vision model that verifies battery component orientation and assembly correctness from images/video, flags misorientation or missing steps in real time, and guides technicians through the correct sequence. Deploy in a US-based/ITAR-compliant manner with on-prem inference, and log each verified step for traceability.

3. Expected Deliverables & Outcomes:

A vision prototype that detects component orientation and assembly-step completion, flags errors in real time, and produces a traceable verification log.

PROBLEM STATEMENT #29 • REF: 6a996c5e239fdeda MANUFACTURING

Automation from Process Data sheet to Mechanical Data sheet

1. Background & Industrial Context:

Mechanical equipment procurement begins with converting process requirements into a Mechanical Data Sheet (MDS) that suppliers can use for design and quotation. Engineers interpret Process Data Sheets (PDS), project specifications, client requirements, applicable codes, materials, and design conditions. Information may be distributed across spreadsheets, documents, drawings, and templates, requiring manual extraction, engineering judgment, and repeated checks. After receiving vendor offers, engineers compare proposed designs with the MDS and enquiry requirements, identify missing information and deviations, request clarifications, and prepare a Technical Bid Evaluation (TBE). The work is repetitive but technically sensitive because overlooked parameters, unit errors, unsupported assumptions, or code mismatches can affect safety, cost, schedule, and equipment performance. Formats vary between projects and suppliers, limiting consistent comparison and traceability. The core need is a controlled method that reduces manual transcription, preserves source references, highlights uncertainty, and helps engineers review requirements and vendor compliance efficiently without replacing accountable engineering decisions.

2. Challenge Statement:

Create an AI-assisted proof of concept for one equipment category that converts a structured or semi-structured Process Data Sheet into a draft Mechanical Data Sheet using an approved template and engineering rules. The POC should extract key parameters, normalize units, identify missing or conflicting information, and link every populated field to its source. Users must review and correct extracted values before approving the MDS. The system should then compare sample vendor offers against the approved MDS, enquiry requirements, material specifications, and selected code rules. It should classify each requirement as compliant, deviating, unclear, or missing; explain the comparison; and generate clarification questions. Finally, it should prepare a draft deviation register and Technical Bid Evaluation with configurable scoring and an auditable decision trail. Use sanitized, synthetic, or mentor-approved documents.

3. Expected Deliverables & Outcomes:

- Input parser for selected PDS, specifications, and vendor-offer formats.
- Draft MDS in a template with normalized units and source traceability.
- Validation report identifying missing, conflicting, or uncertain inputs.
- Vendor compliance matrix classifying requirements as compliant, deviating, unclear, or missing.
- Draft clarification list, deviation register, and Technical Bid Evaluation with configurable scoring.

PROBLEM STATEMENT #30 • REF: 6aab74a163d78afc

ROBOTICS & AUTONOMOUS SYSTEMS

3D mapping and segmentation using any autonomous mobile robot

1. Background & Industrial Context:

Industrial facilities, mining zones, construction sites, and remote infrastructure environments are inherently unstructured, highly dynamic, and spatially vast. Managing, inspecting, and monitoring these assets currently relies on manual field surveys. Inspecting hazardous zones (such as subterranean mines, disaster areas, or complex steel structures) exposes human surveyors to severe safety risks. Conventional surveys generate unstructured raw point clouds or 2D imagery. Converting these raw data feeds into functional digital twins requires manual object segmentation, semantic labeling, and feature extraction--an expensive process that limits real-time spatial analytics.

2. Challenge Statement:

The core challenge requires processing multi-sensor payload feeds (LiDAR, RGB cameras) to perform 3D mapping of the area in a GPS-denied environment . The base system must accurately reconstruct dense 3D environments and classify geometric features and physical assets (e.g., structural pillars, equipment, pathways, fences, humans etc.) without manual labeling. It should work for any kind of terrain and lighting system.

Bonus Capabilities:

1. Integrating full platform autonomy with

a. active exploration (more points for exploratory than following a path)

b. real-time obstacle avoidance

2. Building a dynamic, cloud-synchronized Digital Twin dashboard featuring

a. asset identification

b. automated structural change detection

c. asset condition monitoring

d. spatial querying

3. Expected Deliverables & Outcomes:

Design Process

1. System Architecture: High Level Design and Low Level Design document, etc.

2. Report on: Quantitative benchmark matrix (accuracy, latency, drift) for algorithm selection, dataset details, and runtime compute/power profiling.

Implementation

1. Engineering Report: Challenges encountered, trade-offs, realization vs. target scope, limitations, and comparative evaluation against SOTA baselines.

2. Robot Deployment Package: Plug and play, Hardware agnostic

3. 3D Visualization Viewer: Interactive viewer for spatial mapping, segmented asset rendering, and geometric annotations.

4. Automatics labeling of asset class

PROBLEM STATEMENT #31 • REF: 6a996c5e239fdeda MANUFACTURING

Multipurpose AI-Based Pedestrian Safety System Using RGB and Thermal Sensor Fusion

1. Background & Industrial Context:

People work close to forklifts, trucks, machines, and automated equipment in factories, warehouses, construction sites, mines, ports, and logistics yards. Accidents can happen because the operator cannot see a person in a blind spot or because lighting, dust, smoke, rain, noise, or obstacles make the area difficult to monitor. A person may also enter a restricted zone or move suddenly into the path of equipment. Normal cameras and warning systems can help, but their performance may change with the surroundings and they may miss people or create too many false alarms. Frequent unnecessary warnings can cause operators to ignore important alerts. Different workplaces also have different vehicles, layouts, speeds, safety rules, and levels of connectivity. There is a need for a practical system that can improve awareness of people near moving equipment, identify possible danger early, and give clear warnings while respecting privacy, safety, reliability, cost, and existing work practices.

2. Challenge Statement:

Develop and demonstrate a system that helps prevent collisions between pedestrians and moving equipment in a workplace. Participants may select the setting, users, hazards, information sources, hardware, software, and technical approach. The solution should detect or understand when a person may be in danger, consider the movement of the person and equipment, and provide a timely, clear warning or practical response. It should work under conditions that may include darkness, glare, dust, smoke, rain, blind spots, noise, or partial blockage. Teams should explain how the system handles missed information, false alarms, equipment failure, and situations where it is not confident. The warning should be easy to understand and should not distract or overwhelm the operator. The POC should be tested using safe and realistic scenarios with clear measures such as detection rate, missed cases, false alarms, warning time, range, and response speed. Any automatic control action must be demonstrated safely.

3. Expected Deliverables & Outcomes:

A functional prototype of a multipurpose pedestrian safety system that fuses synchronized RGB and thermal-camera data to detect, in varied operating conditions. Deliverables should include the architecture, sensor-calibration and fusion approach, integration interfaces, safety dashboard, test scenarios, and performance results covering detection accuracy, missed detections, false alarms, operating range, processing latency, environmental robustness, and scalability across different machines and worksites.

PROBLEM STATEMENT #32 • REF: 6a996c60239fdeda SUSTAINABILITY

Current-State Packaging Data and Digital Readiness Assessment for PPWR Compliance Reporting

1. Background & Industrial Context:

Companies selling products in certain regions must provide information about the packaging they use. This may include the material, weight, recycled content, ability to be recycled, unnecessary empty space, restricted substances, suppliers, and supporting documents. In many cases, these details are spread across drawings, spreadsheets, purchasing records, product systems, supplier files, and reports. The same package may be described differently in each place, and important values may be missing, old or difficult to verify. This makes it hard to understand whether packaging meets rules or sustainability goals. It can also delay reporting, increase compliance cost, and risk that products cannot be sold in a market. Teams need a view of what data exists, where it comes from, who owns it, and what gaps must be closed. There is a need for a way to assess packaging information, identify risks, and support better decisions without weakening product protection, customer experience, or affordability.

2. Challenge Statement:

Develop and demonstrate a POC that helps an organization understand its packaging data and readiness for compliance and sustainability reporting. Participants may choose a product group, packaging types, data sources, assessment method, technology, and user experience. The solution should import sample packaging information, compare it with a defined set of requirements, identify missing or conflicting data, and show where supporting evidence is available. It should help users understand packaging material use, recyclability, recycled content, reduction opportunities, supplier information, or other selected factors. Teams should provide clear explanations rather than a simple pass or fail result and allow users to review or correct findings. The POC may use public, synthetic, or approved data and should not claim formal regulatory approval. Results should be measured through data completeness, accuracy, traceability, time saved, and usefulness of recommendations.

3. Expected Deliverables & Outcomes:

Current State Compliance Baseline
PPWR Gap Assessment
Circularity Score Evaluation
Recyclability & Material Optimization
Recycled Content Readiness
Carbon Footprint Reduction Opportunities
Supplier Compliance & Traceability
Packaging Minimization Assessment
Reuse / Refill Feasibility
Compliance Roadmap (2026-2030)

PROBLEM STATEMENT #33 • REF: 6a996c5d239fdeda DIGITAL PLATFORMS

Personalized Streaming Without the Bandwidth Cost

1. Background & Industrial Context:

Streaming companies want to make everything personal -- recommend the right show, insert the right advertisement, deliver the right video quality. The trouble is that personalisation usually means sending more data, and it tends to be needed most in exactly the places where data is scarcest: on a mobile connection, in a small town, or at primetime when everyone is watching at once.

Today, streaming apps handle this crudely. Adaptive bitrate streaming simply watches your connection speed and drops the picture quality when the pipe narrows. It has no idea what is actually on screen: a static news anchor and a fast-moving football match are treated identically, even though one can be compressed hard without anyone noticing and the other cannot.

If the system understood both the content and the viewer, it could spend its limited bandwidth on the parts that matter and quietly degrade the parts nobody notices.

2. Challenge Statement:

Design a solution that delivers a personalised media experience (e.g., adaptive video quality, personalised content assembly, or targeted ad insertion) while minimising bandwidth consumption under constrained or variable network conditions. Approaches may include perceptual or content-aware compression, edge caching, predictive pre-fetching, or AI-driven adaptive encoding that allocates bandwidth to what matters most to the viewer. The primary metric is objective perceptual quality per megabyte delivered, measured against standard adaptive bitrate streaming (HLS or DASH) as the baseline on a common test clip set and network degradation profile, with rebuffering ratio and startup latency as secondary metrics.

3. Expected Deliverables & Outcomes:

A working demo or reproducible simulation showing the proposed approach outperforming the HLS/DASH baseline on objective perceptual quality per megabyte (VMAF or SSIM).

PROBLEM STATEMENT #34 • REF: 6a996c5e239fdeda MANUFACTURING

AI-Vision Powered Smart Inspection System for High-Speed Consumer Electronics PCB Assembly

1. Background & Industrial Context:

Consumer electronics manufacturing demands ultra-high-volume, high-speed production with near-zero defect rates. Surface Mount Technology (SMT) lines producing PCBAs for smartphones, laptops, IoT devices, and wearables operate at speeds of tens of thousands of components per hour, with component sizes shrinking to 0201 (0.25mm x 0.125mm) and below. Defects in solder paste deposition, component placement, and reflow soldering -- such as solder bridges, tombstoning, insufficient solder, component misalignment, and missing components -- must be detected with extremely high accuracy to prevent field failures. Current Automated Optical Inspection (AOI) systems, while fast, rely on rule-based algorithms and golden reference images that generate high false-call rates (5-15%), requiring extensive manual verification and slowing the line. AI-powered deep learning models trained on large defect image datasets can significantly reduce false-call rates while improving true defect detection -- achieving the holy grail of 'zero-escape, zero-false-call' inspection. Additionally, AI can correlate inspection data with upstream process parameters (solder paste volume, placement accuracy, reflow profile) to enable closed-loop process optimization. For LTTS DMS, an AI-vision inspection solution for SMT lines addresses the Tech segment's Consumer Electronics vertical with a high-impact, scalable offering.

2. Challenge Statement:

Develop an AI-Vision Smart Inspection System for consumer electronics PCB assembly that: (a) trains deep learning models (object detection + classification) on PCB defect image datasets to detect and classify common SMT defects (solder bridges, tombstoning, missing components, misalignment, insufficient solder), (b) demonstrates significant reduction in false-call rate compared to rule-based AOI baseline, (c) correlates defect data with upstream process parameters (solder paste volume from SPI, placement accuracy) to identify root causes, (d) provides a real-time defect analytics dashboard with Pareto analysis, trend monitoring, and SPC charts, and (e) proposes a closed-loop feedback mechanism to upstream equipment for process auto-correction.

3. Expected Deliverables & Outcomes:

1. Trained DL defect detection model with precision/recall/F1 benchmarks. 2. False-call rate reduction analysis vs. rule-based AOI baseline. 3. Defect-to-process correlation analytics module. 4. Real-time defect analytics dashboard with SPC integration. 5. Closed-loop process feedback architecture document.

PROBLEM STATEMENT #35 • REF: 6a996c5e239fdeda ENERGY

Sustainable Cooling Solutions for Next-Generation AI Data Centers

1. Background & Industrial Context:

AI systems use powerful computer chips that create a large amount of heat. In data centers, many of these chips are placed close together, so a single equipment rack may need more power and cooling than a normal server rack. If heat is not removed properly, equipment can slow down, fail early, use more electricity, or stop working. Cooling itself can also consume significant power and water, increasing operating cost and environmental impact. The best approach may change with local weather, water availability, electricity price, building design, workload, and plans for future growth. A solution that works well in one location may not suit another. Operators must also keep systems running during maintenance, faults, or sudden changes in computing demand. There is a need for practical cooling ideas that manage high heat safely while balancing energy use, water use, cost, reliability, space, environmental impact, and the ability to expand later.

2. Challenge Statement:

Develop and demonstrate a working physical POC that explores a sustainable way to cool high-density AI computing equipment. Participants may choose the use case, scale, cooling principle, design, materials, hardware, controls, monitoring approach, and testing. The prototype should create or use a heat load and show that the idea can remove or manage heat under representative conditions. Teams should decide which performance measures are most relevant, such as temperature control, energy use, water or fluid use, response time, reliability, cost, size, ease of maintenance, or scalability.

3. Expected Deliverables & Outcomes:

- A safe, working physical cooling prototype using a team-selected approach and scale.
- A suitable heat-load demonstration showing how the concept manages heat.
- Measurements covering selected performance factors such as temperature, energy, water or fluid use, response, reliability, or cost.
- Comparison with a suitable baseline or reference.
- Supporting calculations, simulation, or digital model showing possible scale-up.
- System design and monitoring or control approach.
- Test plan, results, evidence, assumptions, safety measures, operating limits, and known limitations.

PROBLEM STATEMENT #36 • REF: 6a996c5f239fdeda MOBILITY

Real Time ADAS To Detect Parked Vehicle to overtake another vehicle.

1. Background & Industrial Context:

Highway overtaking can become dangerous when the driver's view is blocked by another vehicle, road geometry, roadside infrastructure, poor weather, glare, darkness, or other environmental conditions. Existing driver-assistance systems provide useful information, but their performance can vary with sensing range, object visibility, vehicle speed, traffic behaviour, and the quality or availability of supporting data. A stopped or slow-moving vehicle may be detected too late, confused with another roadside object, or remain hidden until an overtaking manoeuvre has already begun. At highway speeds, even a short delay can significantly reduce the time available to understand the situation, reconsider the manoeuvre, brake, or return safely to the original lane. The challenge is not limited to extending sensor range; it also involves interpreting incomplete and uncertain information, anticipating how a traffic scene may evolve, communicating risk without distracting the driver, and operating responsibly when confidence is low. An innovative approach is needed to improve situational awareness and support safer overtaking decisions while recognizing practical limits, diverse road conditions, human behaviour, and the distinction between driver assistance and autonomous control.

2. Challenge Statement:

Design and demonstrate an innovative proof of concept that improves awareness of hazards and uncertainty during highway overtaking, particularly when the driver or vehicle has a restricted line of sight. Participants may define the target scenario, users, operating conditions, information sources, technical approach, system boundaries, and form of demonstration. Solutions may use onboard, cooperative, connected, infrastructure-based, simulated, or other justified methods, individually or in combination. The POC should interpret relevant traffic conditions, distinguish meaningful hazards from benign objects, consider vehicle motion and available reaction time, and communicate information in a way that supports rather than replaces responsible driver judgement. Teams should determine how their concept responds to occlusion, adverse weather, changing illumination, incomplete data, conflicting observations, and low-confidence situations. Testing should include representative normal, difficult, and failure scenarios, with clear evidence of what the solution can and cannot reliably address. Any physical actuation must be demonstrated only in a safe and controlled environment. The intended outcome is an explainable, testable concept that advances overtaking safety.

3. Expected Deliverables & Outcomes:

- Clearly defined overtaking-safety need, users, scenarios, assumptions, and boundaries.
- Working POC using a participant-selected combination of hardware, software, simulation, connectivity, or infrastructure.
- Method for identifying relevant hazards and assessing risk under restricted visibility.
- Appropriate handling of uncertainty, adverse conditions, missing information, and failures.
- User interaction that supports informed decisions without overstating confidence.
- Evaluation against representative scenarios and justified success measures.

PROBLEM STATEMENT #37 • REF: 6a996c5d239fdeda DIGITAL PLATFORMS

Self-Diagnosing Edge Devices

1. Background & Industrial Context:

Small electronic devices are now installed almost everywhere - inside machines, on poles, under roads, in remote pumping stations. Many of them sit in places where sending an engineer out costs more than the device itself.

The trouble is that these devices tend to fail quietly. A sensor may slowly drift out of calibration, memory may degrade, the board may run hot, or a communication line may fail intermittently. The device eventually either goes silent or, worse, keeps reporting incorrect data that nobody knows to distrust. Somebody then has to travel out, guess at the cause, and usually swap the whole unit.

What is needed is a device that keeps an eye on itself - one that notices its own sensor drifting, its own temperature creeping up, its own power draw looking odd, and says so early. This has to work on a chip with a few kilobytes of memory and a very small power budget.

2. Challenge Statement:

Design and implement an embedded system capable of monitoring its own operational health and autonomously detecting degradation or fault conditions - such as sensor drift, abnormal power draw, communication errors, or thermal anomalies - without relying on continuous human monitoring or cloud connectivity. Faults are to be induced on the running device, by heating the board, injecting noise into a sensor line, or interrupting a bus, and caught while it continues to operate. Teams must state which health signals the device watches, how it identifies anomalies within the memory and compute headroom left on the chosen part, and what it does once a fault is confirmed: log it, degrade gracefully, or raise an alert. Characterise the cost of the self-diagnosis logic as flashed - bytes of flash and RAM occupied, CPU duty cycle, and incremental current draw measured at the supply rail - and justify why it remains viable on a cost-sensitive part. Teams should also explain how diagnostic reports are aggregated across a deployed fleet.

3. Expected Deliverables & Outcomes:

A working prototype on a physical microcontroller or development board, demonstrating live detection of various deliberately induced fault and degradation scenarios. Teams must also report the resource footprint of the self-diagnosis logic (flash, RAM, CPU duty cycle, incremental power draw) and the detection accuracy achieved per fault type.

PROBLEM STATEMENT #38 • REF: 6a996c5d239fdeda DIGITAL PLATFORMS

OTT streaming quality of service

1. Background & Industrial Context:

OTT services deliver video across varied devices, applications, codecs, resolutions, and network conditions. Viewer experience can deteriorate because of slow startup, buffering, frozen or black frames, audio-video misalignment, playback interruption, or ineffective adaptation to available bandwidth. Platform monitoring commonly reports server, application, and network metrics, but these indicators do not always show what a viewer actually sees and hears. Manual observation is time-consuming, subjective, difficult to repeat, and unsuitable for continuous testing across many streams. Quality events may also be reviewed separately from bandwidth, latency, packet loss, and device behavior, making root-cause analysis difficult. As a result, service teams may detect problems late, compare tests inconsistently, or lack clear evidence for prioritizing corrective action. The core need is a repeatable, non-intrusive method that observes playback from the user perspective, measures important quality events consistently, and presents timestamped evidence and comparable results for technical review and meaningful information for corrective action.

2. Challenge Statement:

Create a non-intrusive hardware-and-software proof of concept that monitors selected live or recorded OTT streams from the viewer perspective without modifying the source stream or player. The POC should capture screen output and audio, collect available network or playback telemetry, and identify black frames, frozen frames, startup delay, buffering events, playback interruptions, audio-video synchronization errors, and bitrate or resolution changes. It should correlate detected events with bandwidth, latency, packet loss, and device resource information when available. The system must calculate an explainable Quality of Experience score using configurable thresholds and show how each detected issue affects the score. Results should include timestamps, event duration, confidence, screenshots or clips, and possible contributing factors. Testing should cover a defined set of devices, resolutions, protocols, content samples, and controlled network conditions. The POC is for monitoring and diagnosis by engineering teams; production-scale deployment, exhaustive protocol coverage, and automated service remediation are outside scope.

3. Expected Deliverables & Outcomes:

• Non-intrusive POC that captures selected OTT video, audio, and available telemetry.
• Detection of startup delay, buffering, black or frozen frames, playback interruption, audio-video sync errors, and bitrate or resolution changes.
• Configurable, explainable Quality of Experience score.
• Dashboard showing timestamps, duration, confidence, evidence, and possible contributing factors for each event.
• Test dataset and benchmark report covering defined devices, resolutions, content samples, protocols, and network conditions.
• Source code, setup guide, test evidence, technical documentation, and future-enhancement recommendations.

PROBLEM STATEMENT #39 • REF: 6a996c5f239fdeda MEDTECH

Edge-AI Wearable for Continuous Remote Vital-Sign Monitoring

1. Background & Industrial Context:

Today, if you want your heart rhythm or oxygen levels checked properly, you generally have to go to a hospital. That works if you live near one. Millions of people in rural and semi-urban areas do not. So conditions like an irregular heartbeat, sleep apnea, or diabetic complications quietly build up in the months between one checkup and the next, and are often caught only after real damage has been done.

Fitness bands and smartwatches were meant to help, but most of them are really just sensors with a radio attached. They collect the raw signal and ship it off to a server somewhere to be analysed. That creates three problems: it stops working when the network drops, personal health data travels further than it needs to, and the answer comes back late. In an area with patchy mobile coverage, the device is effectively blind.

The fix is to stop treating the wearable as a dumb sensor and start treating it as a small computer that can spot trouble on its own, warn the patient immediately, and upload the details later whenever a network happens to be available.

2. Challenge Statement:

Design and build a low-cost, low-power wearable prototype that continuously captures at least one physiological signal (e.g., SpO2, ECG, heart rate, or motion) and performs on-device anomaly detection using an embedded AI/ML model. A functional hardware prototype is mandatory. The solution must operate correctly with intermittent or no network connectivity, buffering data locally and syncing opportunistically. The primary metric is detection latency, measured from the onset of the anomaly in the signal to the device raising it, with the false-positive rate over a continuous multi-hour run as the counterweight. Teams may use any combination of low-power microcontrollers, edge-AI accelerators, or TinyML frameworks, and must consider their power budget, sensor accuracy trade-offs, and a realistic caregiver alerting mechanism. Consideration should also be given to data privacy handling.

3. Expected Deliverables & Outcomes:

A functional hardware prototype demonstrating live on-device signal acquisition and anomaly detection, with a recorded or live demo showing at least one detected anomaly. A bill of materials with unit cost, a measured power budget and estimated battery life, and a detection performance summary (latency and false-positive rate).

PROBLEM STATEMENT #40 • REF: 6a996c5f239fdeda MEDTECH

Adaptive AI driven Medical Scan Analysis

1. Background & Industrial Context:

Doctors use medical scans like X-rays, CT scans, and MRI scans to find injuries and diseases. AI can support them by pointing out areas, making measurements, or bringing urgent cases to attention. However, doctors have different levels of experience and may need different kinds of help. A new doctor may value clear explanations and examples, while a specialist may prefer a short summary, measurements, or a second opinion. The support needed can also change with the scan type, patient condition, urgency, and image quality. Most AI tools give similar information to every user and may not fit these changing needs. Poorly designed support can confuse users, create too much trust in the system, or hide uncertainty. There is a need for a safer and more flexible approach that helps each doctor use scan information effectively. The doctor must remain responsible for the final decision, and patient data must remain protected.

2. Challenge Statement:

Develop and demonstrate an AI-based medical scan support system that can adjust how it helps different doctors and clinical situations. Participants may choose a scan type, health condition, user group, data source, and approach. The system could change the amount of detail, explanation, evidence, or guidance it provides based on the user's needs, without making assumptions about a doctor's ability. It should show findings, explain why they may matter, and state its confidence and limits. It should also recognize poor-quality or incomplete scans and ask for review rather than giving a misleading answer. Doctors must be able to question, correct, or reject the system's suggestions. Teams should test whether the doctor and AI working together perform better than either alone, using measures and representative users or simulations. The solution should protect patient privacy, reduce bias, keep a record of decisions, and remain a support tool rather than replace medical judgement.

3. Expected Deliverables & Outcomes:

Machine Learning + Deep Learning + Generative AI driven Adaptive Medical Scan Analysis and Recommendation System

PROBLEM STATEMENT #41 • REF: 6a996c5f239fdeda MANUFACTURING

Predictive Safety on the Factory Floor

1. Background & Industrial Context:

Factories lose money and, occasionally, injure people when machines fail without warning. A bearing wears out, a motor overheats, a pump seizes - production stops, and if someone happened to be standing in the wrong place, it becomes far worse than an expense.

The usual defence is preventive maintenance on a calendar: replace this part every six months whether it needs replacing or not. That is wasteful when the part was perfectly healthy, and useless when the part fails in month four. Real failures do not follow a schedule.

The obvious answer is to monitor machines continuously, but most plant managers assume that means buying new smart equipment, which is far too expensive. So a great many facilities still run twenty-year-old machinery, inspected by someone walking the floor with a clipboard and repaired only after it breaks.

Cheap sensing and on-board intelligence have changed what is possible on old equipment. The open question is whether a retrofit kit can predict a real failure early enough to matter, cheaply enough to scale across a plant, and reliably enough that operators keep paying attention to it.

2. Challenge Statement:

Design and build a low-cost, retrofit-friendly monitoring solution that uses sensor fusion (e.g., vibration, temperature, acoustic, or current draw) combined with an edge-AI or rule-based model to predict impending equipment failure or unsafe operating conditions on industrial machinery. A functional hardware prototype is mandatory, demonstrated on a representative rotating or reciprocating machine - a bench-scale rig such as a motor with an inducible fault is acceptable, provided the fault mode is clearly stated. The solution must attach to existing equipment without major modification or extended downtime. The primary metric is prediction lead time - how far ahead of the failure the warning is raised - with detection accuracy and false-alarm rate as the counterweights. Teams must identify the specific failure mode or hazard targeted and justify sensor selection and placement. Critically, teams must explain how false positives and false negatives are managed, since nuisance alarms destroy operator trust faster than missed faults. A retrofit bill of materials with per-machine installed cost is required to demonstrate viability for budget-constrained facilities.

3. Expected Deliverables & Outcomes:

A functional hardware prototype demonstrated on a representative machine or bench-scale rig, showing prediction of a deliberately induced failure or unsafe condition ahead of occurrence, with a quantified lead time and detection accuracy. Teams must also provide a retrofit bill of materials with per-machine installed cost, and a false-positive / false-negative analysis.

PROBLEM STATEMENT #42 • REF: 6a996c5e239fdeda ENERGY

AI-Driven Battery Intelligence for Predictive Health, Safety, and Lifecycle Optimization

1. Background & Industrial Context:

Battery packs degrade through repeated charging, discharging, temperature variation, high current, storage, and changing usage. Conventional Battery Management Systems measure voltage, current, temperature, and state of charge, but provide limited insight into health, remaining life, or emerging abnormal behavior. Degradation is difficult to assess because capacity loss and internal resistance depend on chemistry, cell history, operating conditions, and measurement quality. Simple thresholds may identify a problem only after performance has deteriorated, while laboratory tests are time-consuming and cannot be repeated continuously in deployed systems. Incorrect health estimates can lead to unexpected loss of range or runtime, unnecessary maintenance, premature replacement, or unsafe operation. Battery data may also contain missing values, sensor noise, inconsistent sampling, and few examples of failures. The core need is a transparent method that converts time-series operating data into reliable health estimates, early warnings, and uncertainty information that engineers can validate before making maintenance or usage decisions.

2. Challenge Statement:

Develop an AI/ML proof of concept for one defined battery chemistry, cell or pack type, and application using public, synthetic data. The POC should preprocess voltage, current, temperature, charge-discharge cycles, capacity, and impedance or operating-condition data. It should estimate State of Health, predict Remaining Useful Life over a defined horizon, and detect selected abnormal thermal or electrical patterns. Every prediction must include confidence or uncertainty and an explanation of the main contributing signals. A dashboard should show battery history, current health, degradation trend, predicted end-of-life range, anomaly alerts, and model performance. Teams should compare the AI approach with simple capacity-threshold or trend-based baselines.

3. Expected Deliverables & Outcomes:

- Defined battery type, use case, data fields, assumptions, and end-of-life criterion.
- State-of-Health estimation model with confidence and explanation.
- Remaining-Useful-Life prediction with uncertainty range.
- Detection model for selected thermal or electrical anomalies.
- Dashboard showing history, degradation, forecasts, alerts, and model metrics.
- Validation report against unseen cells or cycles and simple baselines.

PROBLEM STATEMENT #43 • REF: 6a996c60239fdeda

ROBOTICS & AUTONOMOUS SYSTEMS

Autonomous Vision-Language-Action Physical Systems

1. Background & Industrial Context:

Industrial robots have traditionally been programmed for one fixed job in one fixed place, with parts arriving in exactly the same position every time. That works well for mass repetition, but the moment the product changes, the layout shifts or a new tool is introduced, an engineer has to reprogram the robot - slow and costly. Modern manufacturing needs machines that can understand a situation and adapt. Physical AI aims to close that gap by joining four things in one loop: seeing, understanding, deciding and physically acting. Such a system can take a plain spoken or written instruction, look around using cameras and sensors, work out what the objects are and how they are arranged, plan the steps needed, and carry them out with real hardware. Useful applications include loading and unloading machines, moving material, assembling kits of parts, inspection, sorting and assembly assistance.

2. Challenge Statement:

Build a system that goes from an instruction to a completed physical task as one connected pipeline, rather than a set of separate AI system/sub-system bolted together. Given an instruction, it should identify the relevant objects or work area, understand how they are positioned relative to each other, work out the right sequence of moves, and carry out the task safely using one or more actuators. Solution must cope with normal real-world variation - objects in different positions and orientations, changing light, clutter and obstacles - without being reprogrammed each time. The target is a working prototype, or a simulation, that handles tasks it was not specifically set up for, corrects itself while running, operates safely near people, and clearly outperforms a traditional rule-based setup.

3. Expected Deliverables & Outcomes:

A functional Physical AI robotic prototype (or even simulation) demonstrating natural-language task understanding and autonomous execution. Deliverables should include the physical system / simulation setup, AI models and software, perception-to-action workflow, demonstration of at least one task, performance metrics for accuracy, success rate and response time, and documentation covering system architecture, datasets/training approach, safety mechanisms, limitations, and scalability to industrial applications.

PROBLEM STATEMENT #44 • REF: 6a996c5d239fdeda CYBER SECURITY

Safeguarding Content Rights in the Age of AI-Generated Media

1. Background & Industrial Context:

Media and content companies are investing in AI-powered content generation, across AI-generated advertisements, event highlights, localizations, and synthetic image, video and audio assets. This scale of automated production introduces a significant compliance exposure: content may infringe third-party rights, may misrepresent its own provenance, or may not meet the standards required for public release.

2. Challenge Statement:

Content teams across an organization now use a wide range of generative models and tools, often without a consistent review gate before publication. Design a solution that assesses a media asset prior to deployment and reports on three aspects/dimension:

• Provenance: whether the asset is AI-generated, wholly or partially, including tampering with existing provenance metadata or watermarks.

• Rights exposure: likelihood that the asset reproduces copyrighted, trademarked or licensed material, including likeness and voice.

• Deployment readiness: conformance to a configurable policy set covering brand, regulatory and regional requirements, with a clear go/no-go, flag or block outcome and the evidence behind it.

3. Expected Deliverables & Outcomes:

A working prototype that processes the dimensions, returns a per-dimension confidence score with supporting evidence, and integrates into an existing content workflow. Solutions should state their accuracy and false-positive rate on a declared test set, and should go beyond what standards and invisible watermarking already provide.

PROBLEM STATEMENT #45 • REF: 6a996c5d239fdeda CYBER SECURITY

Governed Autonomous Multi-Agent Investigation System with Secure Tool Use

1. Background & Industrial Context:

A team of AI agents can split a large investigation into smaller jobs, use tools, gather evidence and combine the results. But letting agents act freely brings real engineering and security problems. Two agents may unknowingly repeat the same work. An agent may run far more tool calls than budgeted, use a tool it was never authorised to use, return a conclusion with no evidence behind it, change shared data unexpectedly, or - worst of all - obey malicious instructions hidden inside a document it just fetched. Adding more agents does not automatically help; you can simply get more agents making the same mistake while consuming more time and money. Safe autonomy therefore needs clear job orders, strictly limited powers, necessary tool access, a rule that every claim comes with evidence, controlled permission to change anything, hard stopping conditions, and a complete record of what happened. This challenge is about building a governed team of agents.

2. Challenge Statement:

Build a governed multi-agent system in which a supervisor breaks an investigation into smaller jobs and issues each worker a formal work order defining its goal, success criteria, permitted tools, forbidden actions, resource limits, required evidence and stopping conditions. Each worker returns a standard evidence pack: its claims, the evidence for and against each, its assumptions, its confidence, a log of tools used, and whether it completed the job. Anything a tool returns must be treated as untrusted input and must never change a worker's goal, powers, permissions, budget or safety rules. Reading data and changing data should require different levels of permission. The system must enforce resource budgets, prevent the same data-changing action running twice, handle failures gracefully, and return a partial answer when a job cannot be completed. The supervisor must compare findings and show where workers disagree rather than just picking one, and every order, tool call, decision and stop reason must be recorded in a permanent log.

3. Expected Deliverables & Outcomes:

- One supervisor and two or three bounded worker agents.
- Typed Agent Assignment and EvidenceBundle schemas.
- Three controlled tools with read and action permission separation.
- Per-worker and global tool-call budgets.
- A secure tool gateway and sanitized append-only audit trail.
- Timeout, retry, malformed-result and duplicate-action handling.
- Evidence-based aggregation of worker findings and disagreements.
- Safe stopping with partial results.
- Demonstrations of normal completion, budget exhaustion, injection, unauthorized action, tool failure and conflicting findings.

PROBLEM STATEMENT #46 • REF: 6a996c5d239fdeda CYBER SECURITY

AI-Powered Threat Modeling & Security Validation for Embedded Systems

1. Background & Industrial Context:

Security validation in embedded systems is often:
Done late in lifecycle
Manual and checklist-driven
Not continuously updated with evolving threats

2. Challenge Statement:

Design an AI-assisted threat modeling & security validation tool that:
Automatically identifies attack surfaces from architecture diagrams/code
Generates threat models (STRIDE, attack trees)
Suggests security test cases and validation strategies

3. Expected Deliverables & Outcomes:

Inputs
System architecture diagrams (block diagrams, AUTOSAR XML, etc.)
Source code snippets / interfaces
Known vulnerability databases (e.g., CWE patterns)
Expected Output
Auto-generated threat model
Risk scoring and mitigation recommendations
Security test case suite
Evaluation Criteria
Completeness of threat identification
Relevance to real-world attack scenarios
Integration with V&V lifecycle
Actionability of recommendations

PROBLEM STATEMENT #47 • REF: 6a996c5d239fdeda AI

Evidence-Grounded Divergence and Convergence Engine for Competing Hypotheses

1. Background & Industrial Context:

When a problem is unclear, an AI usually replies with one smooth, confident explanation. Even if you ask for alternatives, it tends to return the same idea reworded. A skilled diagnostician deliberately puts forward genuinely different possible causes, states what each one assumes, looks for evidence that would prove or disprove each, and settles on an answer only when the evidence justifies it. Broadening the search should mean exploring truly different possibilities, not generating more text. Narrowing down should happen because evidence ruled things out, not because one explanation sounded more convincing. The system must also be allowed to keep "none of the above" on the table and to say "I cannot tell yet" when nothing is well supported. A strong reasoning system is one that can explain which options it considered, what evidence changed its mind, what it discarded, and why.

2. Challenge Statement:

Build a reasoning engine that works in two stages. First, it generates a small set of genuinely different candidate explanations for a problem, plus an explicit "unknown" option. They must differ in the cause proposed, the assumptions made, the evidence required or the effects predicted - not merely in wording. For each one, the engine lists what evidence supports it, what contradicts it, what is missing, and which single test would best separate it from the others. Second, it scores the candidates using a documented method and updates those scores as evidence arrives, without counting two pieces of evidence from the same original source as independent confirmation. The engine must end with one of four outcomes: pick the best-supported explanation, combine compatible ones, recommend a specific test, or openly decline to answer. It must report an honest confidence level and work within a fixed exploration budget with a clear rule for when to stop.

3. Expected Deliverables & Outcomes:

- A divergence engine producing substantive hypotheses plus unknown or null.
- A multi-axis method for measuring hypothesis distinctness.
- Structured hypothesis records containing assumptions and predicted effects.
- Supporting, contradicting, missing and dependent-evidence handling.
- A documented hypothesis-scoring and confidence-update method.
- Selection, synthesis, discriminating-test or abstention output.
- A fixed branch budget and stopping rule.
- Calibration and selective-accuracy evaluation on held-out cases.
- Failure tests for duplicate evidence, tied hypotheses and an unlisted cause.

PROBLEM STATEMENT #48 • REF: 6a996c5f239fdeda MOBILITY

Vehicle-to-Vehicle (V2V) Connectivity for Off-highway vehicles

1. Background & Industrial Context:

Off-highway vehicles work in mines, farms, construction sites, ports, warehouses, and areas where routes and rules may change frequently. Machines can have blind spots, and dust, darkness, noise, uneven ground, stockpiles, buildings, or parked equipment can hide vehicles and workers. Operators often depend on mirrors, cameras, horns, radios, warning lights, and their judgement. These methods help, but may not warn early about another vehicle approaching a blind crossing, stopping suddenly, reversing, changing direction, or entering a restricted area. Vehicles may also wait unnecessarily because operators do not know the movement of others. Allowing nearby vehicles to share information could improve awareness, coordination, and safety. Any system must remain useful when location data is inaccurate, messages are delayed, communication is lost, or vehicles use different equipment. There is a need for a reliable and secure approach that supports operators reliably and effectively without distracting them or creating unsafe dependence on connectivity.

2. Challenge Statement:

Develop and demonstrate a POC in which at least two simulated or physical off-highway vehicles share information to improve safety or coordination. Participants may choose the operating environment, vehicle type, communication method, positioning approach, hardware, software, exchanged data, and user interface. The system should turn shared information into a warning, recommendation, or coordination action for a selected scenario, such as a blind crossing, sudden stop, reversing movement, restricted-zone entry, route conflict, queue, or another situation.The POC should be evaluated using suitable measures such as communication range, delay, message success, position accuracy, alert accuracy, response time, security, cost, ease of use.

3. Expected Deliverables & Outcomes:

A functional prototype or simulation demonstrating real-time communication between at least two off-highway vehicle units. The system should include vehicle-data generation or acquisition, wireless data exchange, intelligent detection of unsafe or inefficient operating conditions, and appropriate alerts or recommendations. Teams should demonstrate at least one relevant scenario, such as blind-intersection collision warning, reversing proximity alert, restricted-zone notification, or coordinated vehicle movement. Deliverables should also include the system architecture, demonstration results, performance observations, assumptions, limitations, and a brief roadmap for future scalability and industrial deployment.

PROBLEM STATEMENT #49 • REF: 6a996c5d239fdeda AI

Uncertainty-Aware Causal World Model and Intervention Simulation Engine

1. Background & Industrial Context:

Predicting what usually happens is very different from predicting what will happen if you deliberately change something. A normal prediction model spots patterns in past data - for example, that ice-cream sales and drowning deaths rise together - without understanding what actually causes what. That is fine for forecasting, but useless the moment someone asks "what happens if we turn this valve down?" or "would the machine still have failed if we had replaced that part last month?" To answer those, a model must represent the real mechanisms of a system: its variables, the actions that can be taken, how one state leads to the next, what can be observed, and how much is genuinely uncertain. The aim is not a general model of the world, but one solid model that predicts, compares deliberate actions, reports honest uncertainty, and answers a "what if we had done it differently" question. Such a question is not just another simulation run - it means re-running the same episode with one action changed while keeping everything else about it the same.

2. Challenge Statement:

Build an uncertainty-aware causal model of a system, working against a simulator or similar environment where the true outcomes are known. The model should support four clearly separated abilities: predicting the next state from the current one; predicting the next state given a chosen action; estimating what happens under deliberate interventions, where a variable is forced to a value regardless of what would naturally occur; and answering a "what if" question about a recorded episode by changing a past action while keeping that episode's other conditions fixed. The model may be learned, rule-based or a mix, but it must state its assumptions about causes, what can be observed, and randomness. Every prediction must carry an uncertainty range or a probability spread. Teams should compare predictions against actual outcomes, identify the conditions under which the model is unreliable, and demonstrate that a pattern in the data is not being mistaken for genuine cause and effect.

3. Expected Deliverables & Outcomes:

- A world model implemented against the supplied simulator.
- Documented state, action, observation and intervention definitions.
- Next-state and action-conditioned predictions.
- Predictions for at least two controlled interventions.
- One bounded counterfactual with explicit assumptions.
- Prediction uncertainty or outcome distributions.
- Separate forecasting, action, intervention and counterfactual evaluation.
- A calibration or probabilistic-scoring report.
- Failure demonstrations for low-data conditions, missing observations and observational-interventional disagreement.
- Documentation of unreliable operating regions and model limitations.

PROBLEM STATEMENT #50 • REF: 6a996c5d239fdeda

CYBER SECURITY

Enhancing Cybersecurity Resilience using AI

1. Background & Industrial Context:

Companies use cloud services, online applications, remote access, connected devices, and outside software providers to run daily operations. Each connection can create another way for attackers to steal passwords, enter systems, damage data, or stop important services. Security teams receive large numbers of warnings from different tools, but many are false or repeated. This makes it harder to notice the few warnings that show a real attack. Staff may spend hours collecting information, understanding what happened, and deciding what to do, while an attacker can move quickly. New attacks also change often and may use convincing messages, fake identities, or unfamiliar methods. Automatic responses can save time, but a wrong action might block a genuine user or interrupt business. There is a need for a trustworthy system that helps security teams find important threats, understand the evidence, act faster, and recover safely while keeping people in control of high-impact decisions.

2. Challenge Statement:

Develop and demonstrate an AI-supported cyber-resilience POC for a selected IT environment and a small set of attack scenarios. Participants may choose the users, systems, data sources, threats, AI methods, security tools, and response approach. The solution should help identify important suspicious activity, combine related warnings, explain what may have happened, and suggest or perform suitable actions. High-impact actions must require human approval and all decisions should be recorded. Teams should show how the system handles false alarms, missing or misleading data, uncertain results, and attempts to confuse or misuse the AI. The POC may use public or synthetic data, a safe cyber range, simulated systems, or a suitable combination. It should be compared with a simple rule-based or manual baseline using measures such as detection quality, false alarms, investigation time, response time, recovery time, and analyst effort.

3. Expected Deliverables & Outcomes:

A working prototype demonstrating detection, agentic triage and automated response on a public or synthetic dataset (for example CIC-IDS or a simulated attack range). A system architecture document covering data pipeline, models, agent design and integration with existing SIEM, SOAR and EDR tooling. A live demo of at least two attack scenarios, such as phishing-led credential compromise and lateral movement, from alert through to containment. Quantified results showing reduction in false positives, MTTD and MTTR against a rule-based baseline, with detections mapped to MITRE ATT&CK. A short note on adversarial robustness, AI governance and commercial business value.

PROBLEM STATEMENT #51 • REF: 6a996c5e239fdeda MANUFACTURING

GenAI-Powered Intelligent Manufacturing Knowledge Assistant for Shop Floor Operations

1. Background & Industrial Context:

Manufacturing operations across automotive, aerospace, and commercial vehicle plants generate vast amounts of unstructured knowledge -- Standard Operating Procedures (SOPs), work instructions, maintenance manuals, quality alerts, engineering change notices (ECNs), root cause analysis (RCA) reports, and tribal knowledge held by experienced operators. This knowledge is typically siloed across multiple systems (PLM, QMS, DMS, shared drives) and formats (PDFs, Word documents, legacy databases), making it extremely difficult for shop-floor personnel to find the right information at the right time. Studies show that manufacturing workers spend 20-30% of their time searching for information, leading to productivity losses and delayed problem resolution. Generative AI (GenAI), particularly Retrieval-Augmented Generation (RAG) architectures combined with Large Language Models (LLMs), offers a transformative opportunity to create an intelligent manufacturing knowledge assistant that can ingest, index, and reason over the entire manufacturing knowledge corpus -- answering operator queries in natural language, auto-generating SOPs from engineering specifications, summarizing RCA reports, and providing contextual troubleshooting guidance. For LTTS DMS, a GenAI-powered knowledge assistant is a horizontal accelerator applicable across all Mobility verticals and can be offered as a value-added digital solution to clients.

2. Challenge Statement:

Develop a multimodal manufacturing knowledge assistant for one production line using a governed corpus of SOPs, work instructions, annotated equipment images, quality alerts, and maintenance/RCA reports. The system shall answer operator questions, retrieve procedures from images or text, and draft a change-controlled work instruction. Every statement must cite the exact source document, page/section, image region where applicable, and revision date. It must detect superseded documents, abstain when evidence is missing or conflicting, and route high-risk answers for expert approval. Compare with keyword search and a basic text-only RAG baseline on a fixed expert-authored question set.

3. Expected Deliverables & Outcomes:

1. Versioned multimodal corpus with access controls, metadata, and expert question/answer set.
2. Text, table, and image ingestion with revision and supersession handling.
3. Multimodal retrieval and answer generation with sentence-level citations and confidence.
4. Abstention, conflicting-source detection, freshness checks, and expert escalation.
5. Draft work-instruction workflow with change control and approval.
6. Evaluation versus keyword and text-only RAG using answer correctness, citation precision, retrieval recall, abstention quality, freshness accuracy, latency, and expert time saved.
7. Security and deployment report.

PROBLEM STATEMENT #52 • REF: 6a996c5d239fdeda AI

Unified Multimodal Evidence Representation with Provenance, Temporal Alignment and Conflict Detection

1. Background & Industrial Context:

Real systems produce information in many forms at once: written reports, photographs, tables of readings, and sensor data recorded over time. Most AI tools convert all of this into numeric "embeddings" so it can be searched by similarity. That helps find related content, but it discards information needed in order to trust an answer - which document a fact came from, which exact sentence or table cell, which item it refers to, and at what time. A maintenance report may say a machine ran normally on a particular day while the temperature sensor from that same day shows an overheating spike. A trustworthy system should show both statements and flag the clash, not quietly average them into one meaningless middle value. Linking sources together does not mean squashing everything into a single number - it means connecting them through the same items, events, measurements and timestamps, while every fact still points back to where it came from.

2. Challenge Statement:

Build an engine that reads information arriving in several different forms - such as text, images, tables or sensor readings over time - and turns it into one structured, inspectable picture per case: what is involved, what happened, what was measured, in what units, at what time, and how it all relates. Every fact must be stored with a pointer back to its exact source location, along with when it applies, how confident the system is, and whether it was read directly or worked out. The engine should recognise the same real-world item across different sources even when it is named differently, place events on a shared timeline, and flag a conflict only when two facts genuinely describe the same thing at the same time in comparable terms. Conflicts must stay visible, never averaged away. Teams should check their confidence scores against known answers and show how the engine behaves when a source is missing, a name is ambiguous, units differ, or two sources disagree.

3. Expected Deliverables & Outcomes:

- A working engine processing text, image and structured or time-series data.
- A unified state schema containing entities, events, attributes and temporal context.
- A ClaimRecord schema separating extracted and inferred claims.
- Claim-level evidence pointers for all supported modalities.
- Cross-modal entity linking, timeline alignment and conflict detection.
- Confidence reporting with accuracy and calibration evaluation.
- An interface for inspecting claims and supporting evidence.
- Demonstrations covering missing data, entity ambiguity and cross-modal disagreement.

PROBLEM STATEMENT #53 • REF: 6a996c5d239fdeda AI

Efficient and User-Controllable Context Memory for Long-Running Agentic AI

1. Background & Industrial Context:

A language model reads everything it is given as "tokens" - roughly, pieces of words - and every token costs money and time. An agent working with the same user over weeks or months builds up a long history of conversations, documents and tool results. Most systems send all of it back to the model every time, making replies slow and expensive, until eventually the history no longer fits the model's context window and material is cut off at random. Shrinking the history is the obvious fix, but cut too much and the agent forgets preferences, drifts from the original objective and answers wrongly. A second problem compounds the first: none of this is visible. Users cannot see what the agent is remembering, cannot tell when it went off track, and can only start again. Combining efficient compression with a visible, versioned history would make long-running agents both affordable and trustworthy.

2. Challenge Statement:

Build a context memory system for a long-running AI agent that is both efficient and transparent. The system should decide automatically what must be kept word-for-word, what can be reduced to a short summary, and what can safely be discarded, working across hundreds or thousands of messages spanning many sessions without depending on one specific model. Every such decision must be recorded rather than hidden: the memory should be saved as a chain of versioned steps, each labelled with the topic being worked on, so the user can see what is currently held, trace how the agent's understanding evolved, restart the task from any earlier point, branch off to try a different direction, and merge branches back together. Users should also be able to trim the context down or expand it again by hand, and the system should explain what it kept, what it dropped, and why. Success means a measurable reduction in tokens, cost and response time, with answer quality holding steady or improving, and with the user able to inspect and correct the agent's memory at any point instead of restarting.

3. Expected Deliverables & Outcomes:

A working proof-of-concept demonstrating efficient, user-visible context memory across a long multi-session conversation. Deliverables should include: (1) an architecture covering memory storage, compression, retrieval and context reconstruction; (2) implementation of the versioned context chain with trace-back, restart, branch and merge; (3) an interface for inspecting, trimming and expanding context, with explanations of what was kept or dropped; (4) an evaluation comparing baseline and optimised systems on tokens, cost, latency and answer quality; (5) a demonstration of recovering from drift by returning to an earlier context version.

PROBLEM STATEMENT #54 • REF: 6a996c5d239fdeda AI

Safe Feedback Driven Continuous Improvement of Agentic AI Systems Without Model Retraining

1. Background & Industrial Context:

AI "agents" are software helpers that can plan, use tools and take decisions on their own. When people use them, they naturally say things like "this answer was wrong" or "this was helpful". Today that feedback is mostly thrown away, because the only known way to make the AI better is to re-train the large language model behind it - slow, costly and demanding special hardware. So the agent keeps repeating the same mistakes. There is another way: instead of changing the model, change what surrounds it - its memory, its instructions, the way it picks tools, the steps it follows. But feedback cannot be trusted blindly. Users can be wrong, biased, or contradict each other, and a careless update can leave the agent worse than before.

2. Challenge Statement:

Build a system that lets an AI agent get better with use - without touching or re-training the model itself. Collect feedback in two ways: what users say directly and what users do (accepted the answer, redo of the task, etc). Then use that feedback to update the agent's memory, instructions, tool choices or workflow steps so the next attempt is better. This involves judging the feedback to decide which feedback is genuine and useful, and which is harmful - and reject the bad ones. It must ensure that fixing one task does not break another.

3. Expected Deliverables & Outcomes:

Participants are expected to deliver a working proof-of-concept demonstrating feedback-driven improvement of an AI agent over multiple interactions. Deliverables should include:
(1) a detailed system architecture describing feedback collection, storage, evaluation, and adaptation mechanisms
(2) implementation of one or more feedback learning strategies that do not require model retraining
(3) safety mechanisms to detect and prevent performance degradation caused by poor-quality feedback
(4) an evaluation framework comparing agent performance before and after feedback incorporation using measurable metrics such as task success rate, accuracy, user satisfaction, latency, and consistency
(5) a demonstration showing how the agent progressively improves on representative tasks.
Additional consideration will be given to solutions that support autonomous feedback processing, explainability, trustworthiness, multi-user environments, and continuous learning.

PROBLEM STATEMENT #55 • REF: 6a996c5d239fdeda AI

AI reviewing AI - AI Assisted Continuous Review and Performance Improvement of AI Agents

1. Background & Industrial Context:

Once an AI agent is running in a real workflow, it needs to be continuously be checked on whether its answers are still correct, relevant, safe and consistent. Normally that checking is done by humans reviewing outputs, or by running a fixed set of test questions. Both are slow, expensive and cannot keep pace with a system that changes every week. An alternative is to use a second AI as the reviewer, reading the first AI's answer, its reasoning steps and the tools it used, then scoring it and suggesting fixes. The catch is that the reviewer is also an AI, so it can be biased, inconsistent, invent criticism, or simply prefer answers written in its own style. Its verdict therefore cannot be treated as the truth and acted on automatically.

2. Challenge Statement:

Build a setup where one or more "reviewer" AIs continuously grade another AI agent and suggest improvements. The reviewer should look not only at the final answer but, where possible, at how the agent planned, searched, which tool it chose, whether the task actually finished. Accepted feedback can then be used to improve the agent's instructions, its search strategy, its memory, its workflow, its tool routing or its safety rules. Crucially, the reviewer must not be allowed to change the agent directly. Every suggested change has to pass tests, confidence thresholds and a re-check of older tasks before it is switched on.

3. Expected Deliverables & Outcomes:

Expected to deliver a working proof of concept demonstrating AI-driven evaluation and controlled improvement of an AI model or agent. Deliverables should include:
(1) a system architecture covering the target agent, reviewer models, feedback store, improvement engine, validation gate, monitoring, and rollback
(2) implementation of single-reviewer or multi-reviewer evaluation
(3) mechanisms for transforming accepted reviews into changes to prompts, retrieval, memory, workflows, tools, or guardrails
(4) an evaluation framework comparing baseline and improved performance
(5) experiments measuring review accuracy, performance gains, latency, and cost.
Teams should submit a report describing methodology, datasets, evaluation criteria, experimental results, reviewer failure modes, limitations, governance requirements, and production recommendations.

PROBLEM STATEMENT #56 • REF: 6a996c5d239fdeda AI

Agent Control Tower -- Multi-Agent Orchestration Dashboard

1. Background & Industrial Context:

Enterprise teams running multiple AI agents across departments (finance, IT, HR, support) have no unified visibility into agent activity, decisions, failures, or costs. Blind spots lead to runaway costs, silent failures, and compliance risks.

2. Challenge Statement:

Build an AI-powered Agent Control Tower that provides real-time orchestration, monitoring, and governance of multiple autonomous AI agents running across an organization.

1. Agents from different frameworks (LangChain, AutoGen, CrewAI) emit logs in inconsistent formats
2. No single pane of glass for agent health, task progress, and cost attribution
3. Human-in-the-loop escalation has no standardized trigger mechanism
4. Agent dependency chains create cascading failures that are hard to trace

3. Expected Deliverables & Outcomes:

1. Unified dashboard showing all active agents, their status, tasks, and cost per run
2. Agent lineage graph -- visual DAG of agent-to-agent calls and tool usage
3. Anomaly alerts -- flag agents consuming excessive tokens, looping, or failing silently
4. Human escalation interface -- approve/reject agent decisions in real time
5. Cost attribution -- per-agent, per-team, per-task spend breakdown

Replay & audit log -- step-by-step agent decision trace for compliance

PROBLEM STATEMENT #57 • REF: 6a996c60239fdeda **SUSTAINABILITY**

Zero-Waste Grocery - An Agentic AI Approach to Perishable Demand & Shelf-Life Management

1. Background & Industrial Context:

Perishable grocery categories such as produce, dairy, bakery, and meat have variable shelf lives. Retailers must balance product availability against spoilage while responding to demand, delivery schedules, promotions, weather, events, and customer behavior. Decisions are made across store operations, category management, replenishment, and supply-chain teams using data from point-of-sale, inventory, warehouse, pricing, and waste systems. These sources may be delayed, incomplete, or disconnected, making it difficult to understand remaining shelf life and likely demand for each item at each location. Excess stock leads to markdowns, waste, and lost margin, while insufficient stock causes missed sales and poor customer experience. Existing planning methods may rely on historical averages and fixed rules that cannot respond quickly to local changes. The core need is timely, coordinated decision support that combines demand and freshness information, identifies emerging risks, and helps planners choose replenishment, markdown, or redistribution actions before products expire or shelves become empty.

2. Challenge Statement:

Develop an agentic AI proof of concept for selected perishable categories in a simulated grocery network. The system should ingest historical sales, inventory, receipts, prices, promotions, waste, product shelf-life, weather, and local-event data. It should forecast short-term demand, estimate remaining shelf life and spoilage risk, and detect likely stockouts or excess stock. Separate agents may assess demand, freshness, replenishment, markdown, and inter-store transfer options. An orchestration component must compare these options, apply business rules, and propose one action for each affected SKU and store. Every recommendation should state timing, quantity or markdown level, confidence, expected effect, and supporting evidence. A planner must be able to modify, approve, or reject the proposal before action. Use synthetic, public et all and simulated retail-system interfaces.

3. Expected Deliverables & Outcomes:

• Prepared dataset and ingestion workflow for selected sales, inventory, shelf-life, promotion, waste, weather, and event data.
• Demand-forecasting and spoilage-risk models for selected SKUs and stores.
• Multi-agent recommendation engine with orchestration and configurable business rules.
• Planner dashboard for reviewing, modifying, approving, or rejecting proposed actions.
• Simulated ERP, WMS, and POS interfaces with decision and feedback logs.
• Evaluation report comparing the POC with simple forecasting and rule-based approaches.

PROBLEM STATEMENT #58 • REF: 6a996c5d239fdeda DIGITAL PLATFORMS

Develop a responsible social media application

1. Background & Industrial Context:

Social media helps people learn, share ideas, follow interests, and stay connected. However, users often have little control over what appears in their feeds. Someone looking for science, sports, art, technology, or personal stories may instead see unrelated posts, advertisements, shocking content, or artificial trends. Platforms often recommend content to keep people watching, rather than to support what they want to do at that moment. Users may also face bullying, false information, fake accounts, hidden promotions, and content created or changed by AI without clear labels. Children and other vulnerable users can face risks. At the same time, strict filtering can hide different opinions and limit useful discovery. There is a need for a more responsible social media experience that gives people choice without secretly profiling them. It should make online participation safer, clearer, and more useful while protecting privacy, free expression, creativity, and the ability to connect with communities.

2. Challenge Statement:

Develop and demonstrate a social media experience that gives users more control over what they see and how they interact online. Users should be able to choose, combine, or change interests such as learning, sports, art, technology, entertainment, personal stories, or social causes. The experience should reduce unwanted distractions while still allowing people to discover different ideas when they choose. It should also help users create content together safely for a shared topic, event, or purpose. Participants may decide how contributions are invited, reviewed, arranged, published, and reused. The solution should explore ways to handle harmful content, false information, fake accounts, artificial popularity, hidden advertising, and AI-created or AI-changed material. It should protect personal information, consider child safety, explain decisions, and allow users to report or question content. Teams may choose their features, design, technology, and approach, but should show how the idea improves user control, trust, safety, and usefulness.

3. Expected Deliverables & Outcomes:

A functional prototype of an interest-focused social platform where users can select, combine, and switch between Interest Modes and Focus Modes. The prototype should demonstrate relevant content feeds, interest-based communities, and moderator-controlled video stitching that combines approved contributions into one shared creation. It should include AI-content labeling, harmful-content moderation, bot and manipulated-engagement detection, child-safety controls, privacy and audience settings, and contributor consent. Deliverables should also include the proposed modular architecture, secure APIs, scalability approach, administrator dashboard, and demonstration of key user journeys.

PROBLEM STATEMENT #59 • REF: 6a996c5d239fdeda AI

Cost Optimisation using LLM routing

1. Background & Industrial Context:

Every request sent to a large language model costs money, charged by the amount of text going in and coming out. As organisations roll out AI across many teams, this bill grows quickly and is becoming a serious concern. Much of the waste comes from using one large, expensive model for everything, including simple jobs such as reformatting a list, extracting a date or classifying a message, which a smaller and cheaper model, or one running on the organisation's own hardware, could handle just as well. The expensive model is genuinely needed only for tasks that require deep reasoning. If a system could assess each incoming request, judge how difficult it really is, and send it to the cheapest model capable of doing it properly, costs could fall sharply with no loss of quality. The difficulty lies in judging correctly: send a hard question to a weak model and the answer is wrong; send an easy one to the strongest model and money is simply wasted.

2. Challenge Statement:

Build a router that sits in front of several language models of differing cost and capability, including at least one running locally, and decides which should handle each incoming request based on how complex the task is and how much reasoning it needs. Where a request has several parts, the router should be able to split it, send each part to the most suitable model, and combine the results into one answer. The goal is a measurable reduction in cost while keeping output quality comparable to always using the strongest model. The router must be transparent and controllable: users or administrators should be able to see which model handled what and why, and be able to override the choice when needed.

3. Expected Deliverables & Outcomes:

transparency and control of LLM routing.
trade off balancing system cost vs output quality
benchmarking machanism to measure the cost reduction

PROBLEM STATEMENT #60 • REF: 6a996c5d239fdeda AI

Trust-Aware AI Agents - Confidence Estimation for Reliable Decision Making

1. Background & Industrial Context:

AI systems are now used in places where a wrong decision costs money, breaks equipment or puts people at risk. Language models reason impressively well, but they have one weakness: they do not know when they do not know. They answer a question they have no information about in exactly the same confident tone as one they are sure about. A human expert behaves differently - before acting, they judge how sure they are, and if unsure they look something up, ask a colleague, or escalate. Giving AI that same sense of self-doubt remains an unsolved problem. An agent that could reliably distinguish "I am confident" from "I am guessing" would be far safer to trust with real work.

2. Challenge Statement:

Build an AI agent that estimates how confident it is in its own answers, reasoning and planned actions - and then acts sensibly on that number. When confidence is high it should just answer. When confidence is low it should do something else: search for more information, run a tool to verify, ask the user a clarifying question, hand over to a specialist agent, or escalate to a human. The confidence score must be explainable - the user should be able to see why the agent was unsure.

3. Expected Deliverables & Outcomes:

Expected to deliver a working proof-of-concept demonstrating confidence aware decision making within an AI agent or multi-agent system. Deliverables should include:
(1) A detailed architecture covering confidence estimation, uncertainty detection, verification mechanisms, escalation logic, and monitoring components.
(2)Implementation of one or more confidence-scoring approaches.
(3)Mechanisms for automated decision routing based on confidence levels.
(4)Evaluation comparing traditional agents against confidence-aware agents. Experiments demonstrating reduction in hallucinations, failed decisions, and unnecessary escalations.
A live demonstration showing confidence-driven behavior across multiple scenarios.